Sustainability & Climate Justice Commission Agenda

Date	Monday, January 12th
Time	5 pm
Location	City Hall Council Chambers
Watch Online	https://www.youtube.com/@CityofIthacaPublicMeetings

Item	Voting	Presenter	Time Allotted
1. Call to Order 1.1 Agenda Review	 No	 David Kay, Chair	 5
2. Public Comment 2.1 Statement from the Public 2.2 Commission Response	 No No	 David Kay, Chair	 5 5
3. Announcements, Reports & Presentations 3.1 Updates 3.1.1 Sustainability Planner	 No	 Rebecca Evans, Director of Sustainability	 5
4. New Business 4.1 Climate Action Plan Review 4.2 2026 Commission Planning	 No Yes	 Rebecca Evans	 30 25
5. Meeting Wrap-up 5.1 Next Meeting: February 9th 5.2 Agenda Planning for Next Meeting 5.3 Adjournment	 No No Yes	 David Kay, Chair	 5

Executive Summary

Communities across the country are facing rising challenges from severe weather, changing energy demands, and growing economic pressures. In 2019, the City of Ithaca adopted the Ithaca Green New Deal, a community-wide plan to achieve carbon neutrality by 2030 while strengthening economic opportunity, public health, and overall community resilience.

Achieving these goals requires a coordinated, whole-of-community effort. Most of Ithaca's emissions result from energy use in buildings, transportation, and power generation. Reducing emissions will depend on both lowering overall energy demand and shifting toward cleaner, more reliable energy sources. At the same time, an emissions-only approach could overlook pressing community challenges – such as housing affordability, workforce stability, or public safety – services critical to municipal operations and community wellbeing. Ithaca's plan therefore adopts a "net-damages" approach, balancing reductions in greenhouse gas emissions with proactive strategies to protect welfare, safeguard resources, and build resilience in ways that can be easily integrated into existing municipal budgets and processes. The proposed plan is intentionally flexible and designed to be interactive for policymakers and the public. We encourage the community to engage with the included matrix, as the recommendations vary in cost, impact, and duration.

To ensure alignment with the City's Comprehensive Plan, downtown development strategies, and legislative priorities, the Climate Action Plan is organized into six central focus areas:

- **Housing –** Expand access to affordable, secure, and climate resilient housing so residents can live in stability and dignity.

- **Labor –** Support workforce development, create pathways to quality local jobs, and encourage innovation in clean energy and business models while protecting workers.
- **Public Health –** Strengthen community health through preventative and adaptive measures that address risks tied to a changing climate.
- **Equity –** Ensure that strategies are fair and inclusive, recognizing that certain populations face disproportionate risks from both climate impacts and the transition.
- **Power Reliability – Guarantee** continuous, affordable, and resilient access to electricity to sustain hospitals, schools, businesses, and homes, while advancing local energy independence.
- **Emergency Preparedness –** Enhance readiness for flooding, extreme heat, and other hazards by integrating prevention, response, and recovery strategies.

By anchoring climate action in these familiar and practical areas, the City seeks to achieve:

- Greater transparency and accountability
- Alignment with existing community plans and priorities
- A fair distribution of responsibility and benefits
- Broad-based public support across political and community lines

Confronting challenges of this scale requires a unifying approach. Too often, climate and sustainability initiatives are confined to specialized staff or limited budgets, significantly slowing progress. By embedding climate action into the same areas that matter to most residents – housing, jobs, health, and safety – Ithaca can broaden participation, reduce friction, and ensure steady progress.

This strategy represents a shift in practice: moving away from climate action as a separate agenda and instead treating it as an integrated pathway toward stronger infrastructure, healthier communities, and a more stable economy. By addressing immediate needs while reducing long-term risks, Ithaca positions itself to advance what we call **Dignity Toward Decarbonization –** an approach that unites environmental responsibility with human well-being, economic stability, and community resilience.

Baseline Assessments

The Ithaca Green New Deal

On June 5th, 2019, the City of Ithaca Common Council unanimously adopted the Ithaca Green New Deal (IGND) resolution, a government-led commitment to community-wide carbon neutrality by 2030 that focuses on addressing historical inequities, economic inequality, and social justice. Two years after the resolution was sign, Ithaca established itself as a world-leader in climate mitigation planning and continues to pave the path forward as a blueprint for other cities across the U.S. and the globe.

The IGND Goals

The 2019 resolution laid out four explicit goals for City government:

1. Reach community-wide carbon neutrality by 2030.
2. Ensure the benefits of the Ithaca Green New Deal are shared among all local communities to reduce historical social and economic inequities.
3. Meet the electricity needs of government operations with 100% renewable electricity by 2025.
4. Reduce emissions from the City vehicle fleet by 50% by 2025.

The City has achieved many important milestones of progress on the IGND since 2019, including goal #3, many of which provide critical waypoints in the development of the Climate Action Plan.

The City of Ithaca

Around 56% of the total 32,000 residents in Ithaca are between the ages of fifteen and twenty-nine. Cornell University itself occupies the northeast area of Ithaca, and students tend to live in neighborhoods of close proximity, such as Collegetown, Cornell Heights, and University Hill. In contrast, the western neighborhoods of Fall Creek, Southside, and West Hill have a higher concentration of long-term residents.

The current 32,000 resident population in Ithaca is the result of slow but continual growth for several decades. Recent census data even suggests a significant spike after the COVID shutdowns in 2020, where the population grew by 2,000 residents between 2019 and 2021, which is as much growth as between 2019 and 2000. This spike may be attributed to the increase in working from home opportunities, which census data suggests also spiked significantly between 2019 and 2021.

Ithaca’s large student population has a significant influence in Ithaca’s demographics. The transient nature of students means fewer people remain in Ithaca long enough to get married or establish families. Of the almost 14,000 total households in Ithaca, over 70% identify as single, and around 22% as being a ‘married couple’. Contrast that with the whole of New York State’s census data, where the difference between single households and married couples was 50% and 43% of the total household population respectively.

Ithaca is a majority-white city, with around 64% of the population identifying as such in the 2023 census. There is a sizable Asian and African American population who make up around 17% and 6.5%, respectively. Identity-based volunteer organizations to organize events and strengthen minority voices in the community.

Housing

Housing affordability, access, and quality are concerns for both current residents and those who would like to live within the city. The cost of housing in Ithaca has increased dramatically over the past two decades, with the median home value rising 56% since 2014 and median rent up 46% during the same period[1]. Extraordinarily low vacancy rates for both rental and homeowner units suggest that a lack of supply is a primary factor fueling these increases. The large student population within the city and the additional pressure it places on the rental housing market also have a direct impact on the availability, quality, and affordability of rental units. For households with lower incomes, this constricted market leaves very few options for decent housing that is both affordable and conveniently located. Renters make up more than 70% of the city's population, and well over half of this group are considered housing burdened, paying

Labor

The presence of higher education institutions in and around Ithaca have an overwhelming influence on the city's labor demographics. Regarding the 2023 census profile, the 'Educational services, and health care and social assistance' sectors account for over 54% of Ithaca's working population. The next highest sectors are arts/entertainment, which include theater organizations, sports facilities, museums, and other recreation activities, and science/management, which include highly skilled professions in science, law, and engineering, accounting for 12% and 11% respectively.

When viewing economic and labor statistics in Ithaca, it is important to contextualize the disproportionate influence college students have on census data. For example, Ithaca's employment rate of 50% may seem shockingly low, but that's not including the 46% of residents marked as 'not in labor force' which typically applies to young adults and college students working toward their degrees. This also throws off income data, as the median income for Ithaca is reported to be $48,617 is brought down by the 17% of residents reporting a household income under $10,000/year. The median income would likely be higher if not for the college students who are not in the labor force. Contrast the individual

[1] U.S. Census Bureau. 2000 Census vs. 2008-2012 American Community Survey (ACS).

median income with the median income for a family in Ithaca, which is reported to be $122,065.

A Tompkins County-led study in 2025 calculated what a living wage for a single adult in Tompkins County would look like. This included spending data for nine basic needs categories to generate an annual basic needs budget, divided by an hourly wage for a full-time worker over the course of a year. The study concluded that a 2025 living wage would be $24.82/hour. This is a 34.5% increase from 2023's findings, where the living wage was calculated to be $18.45. The current minimum wage applicable to Tompkins County is currently $15.50, creating a gap of $9.32. The study also determined that nearly half of all workers in Tompkins County make less than the $24.82/hour threshold of a living wage, and this number disproportionally applies to women and people of color.

Of the roughly 29,400 working population in Ithaca, around 15,000, or over half, are not regarded as "workers in the area" meaning they commute from out of town. For people working and living in Ithaca, the number of people who walk to work, take personal transportation, and work from home each account for around 25% of the total labor force. The 26% of workers who work from home is, unsurprisingly, a sharp increase from pre-2020 lockdowns, with the 2019 census reporting only 9% of workers working from home.

Equity

Public Health

Emergency Response

Power Reliability

Climate Trends & Vulnerabilities

Climate Risks & Interdependencies

Current societal systems and infrastructure are highly interdependent. Complex systems like housing access and career mobility are complicated and compounded by race and class. Similarly, infrastructure we use every day like telecommunications, energy distribution, and wastewater structures rely heavily on each other to meet our existing and future needs. Even further, social, economic, and physical systems are constantly interacting, creating a web of relationships that are intimately connected, ultimately affecting access, affordability, and stability of each other independently and together.

Climate instability, which frequently results in events like heat waves, heavy rainfall, frequent temperature fluctuations, and flooding, can deeply disrupt the functioning of these structures, causing rippling impacts across systems. When these system vulnerabilities are stacked, we experience what is called “cascading risks” or logical interdependencies, where failure in one system causes cascading impacts on the economy, society, and the environment for an individual or for an entire population.

Climate Trends In 2024, the New York Academic of Sciences published state-specific climate projections as part of the New York State Climate Impacts Assessment. This section summarizes the key findings noted in the study, titled *New York State’s Changing Climate*[1] and is provided below.*Key Finding 1: Average and maximum temperatures have increased in New York State since the early 20th century and are projected to continue to rise throughout the 21st century.* The state has warmed more rapidly than the national average, and winter is warming more rapidly than other seasons. Heat waves are expected to occur more often and become more intense, posing greater risks for human health, built infrastructure, ecosystems, and other sectors. New York City is projected to remain the warmest part of the state; northern regions will continue to be relatively cooler while still experiencing large increases in temperature and extreme heat.

		Baseline	2030s 25th percentile	2030s 75th percentile	2050s 25th percentile	2050s 75th percentile	2080s 25th percentile	2080s 75th percentile
Northern	Canton	93	99	103	104	109	107	120
	Dannemora	93	99	103	103	109	107	119
	Lake Placid	86	91	95	95	100	98	110
	Watertown	91	98	102	102	109	106	121
Central and Western	Binghamton	94	100	104	104	110	107	120
	Buffalo	95	102	106	106	112	110	125
	Cooperstown	92	98	102	102	108	105	119
	Elmira	95	101	104	104	110	107	120
	Fredonia	89	96	99	100	105	104	118
	Norwich	92	98	102	103	108	105	118
	Oswego	90	97	103	103	110	107	125
	Rochester	96	103	106	107	112	110	123
	Syracuse	96	102	107	107	112	110	124
Southern and Eastern	Albany	99	106	110	110	117	113	128
	Bridgehampton	92	98	103	103	109	107	122
	Central Park	101	107	112	112	119	115	130
	Dobbs Ferry	100	107	111	112	118	115	130
	Mohonk Lake	99	105	109	110	116	112	127
	Saratoga Springs	97	104	108	108	114	111	126

Extreme danger (≥124°F)
Danger (104°F–124°F)
Extreme caution (91°F–103°F)
Caution (80°F–90°F)

Key Finding 2: New York State has experienced increases in total precipitation and heavy precipitation events, and these trends will continue through the end of this century. Heavy rainstorms that lead to flooding are projected to become more frequent across the state.

Precipitation is expected to increase the most in winter. Lake-effect snowfall is projected to increase over the next few decades, but as temperatures continue to rise, more winter precipitation near the Great Lakes will fall as rain. Elsewhere in the state, snowfall and snowpack are likely to decrease with warmer winter temperatures.

Key Finding 3: Climate change is creating conditions that will increase the frequency and severity of many types of extreme events. Several types of storms are expected to become more intense, with heavier rainfall, stronger winds, and higher storm surge along the coast driven by sea level rise. Short-term summer droughts could increase due to changing precipitation patterns and increased temperatures. Wildfires are unlikely to become much more common within New York State due to climate change, but air quality impacts from large fires elsewhere in North America could increase in the future.

Key Finding 4: Sea surface temperature, sea level, and coastal flooding are increasing along New York State's coast. Sea surface temperatures are rising more rapidly in the state than the global average. Sea level along New York's coastline has risen almost 1 foot in the past century and is projected to increase by another 1–2 feet by mid-century, making chronic flooding more common in low-lying coastal neighborhoods. Ocean water is also becoming more acidic as it absorbs excess carbon dioxide from the atmosphere, although stormwater runoff currently has a larger effect on acidity in New York's coastal waters.

Key Finding 5: New York State's lakes and rivers have experienced increased water temperature, fluctuating water levels, and decreased ice cover, and these changes are expected to intensify in a warmer, wetter future. Lakes are projected to experience more severe summer heat waves and decreased winter ice cover as temperatures rise in the coming decades. The Great Lakes could experience greater year-to-year variability in water levels, driven by periods of drought and extreme precipitation. Flood intensity and damages are expected to increase with extreme rainfall and broader changes in streamflow.

Climate Vulnerabilities

The following section represents community vulnerabilities to climate change supported by data and research that is currently available and accessible. The risks described in each section are not exhaustive, but are an attempt to reflect some of the more pervasive vulnerabilities that create systems of cascading risk. It should also be noted that localized data are not readily available for all sectors or vulnerabilities. Therefore, staff recommend exploring partnerships that will enable the collection and development of these datasets to facilitate benchmarking and progress tracking.

Housing

The City of Ithaca's housing stock exhibits multiple vulnerabilities with climate impacts exposure. Namely, concern exists with chronic and acute flooding incidents, extreme heat, real estate market property valuations, and risk associated with natural disasters. Hazards related to each risk scenario are described in more detail below.

As described in the *Climate Trends & Vulnerabilities* section, the City of Ithaca can expect a multitude of changing conditions that may make the city and its residents more susceptible and vulnerable to climate-influenced disasters like flooding, urban fires, extreme heat, and other extreme weather events. While immediate preparedness and response protocols are essential to keeping our community safe during and immediately after catastrophic events, it is equally important to understand how weather events can impact other sectors of our society and how gaps in planning can influence vulnerability.

Chronic & Acute Flooding

In 2022, and again in 2023, the Federal Emergency Management Agency (FEMA) published preliminary maps indicating the severity of inundation during a 1-in-100-year flood event. In theory, this represents a 1% chance of severe flooding in any given calendar year, however, the actual flood risk is much more likely, with a 26% chance of flooding over the life of a 30-year mortgage[2]. Today, a flooding event like this could affect 2,573 properties in the City of Ithaca, resulting in significant property and infrastructure damage[3]. The extent of that damage could vary from difficult-to-treat mold infestations to more pervasive structural damage or, worse, to total property loss.

Even when a building remains standing after a flood, flooding can still cause significant structural damage to a home, resulting in costly repairs and safety hazards. Sometimes, this damage includes buckling floorboards, cracks in the foundation, or frayed and damaged electrical wires and components. Most often, however, damage comes in the form of mold and mildew, presenting significant health risks to building occupants. Mold is known to grow on nearly all surfaces found in homes and growth can start on damp surfaces within 24 to 48 hours, reproducing quickly by spreading their spores through the air[4]. As mold colonies proliferate, they become increasingly difficult to control and human health risks increase, causing respiratory problems like asthma, allergies, and infections. People like young children and infants, the elderly, and those with chronic illnesses or weakened immune systems are at even greater risk of acquiring severe illness from mold exposure.

In the latter scenario, when flooding causes more pervasive structural damage, housing in the public and private sectors run significant financial risk after a major flood event. For example, in 2021, severe thunderstorms, torrential rainfall, and hail affected many regions in Germany and resulted in devastating flooding. The infrastructure damage both stalled

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[2] First Street Foundation. (2021). First Street Community Risk Data V1.3 (1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.5711172

[3] First Street Foundation. (2021). First Street Foundation Property Level Flood Risk Statistics V1.3. (1.3). [Data set]. Zenodo. https://doi.org/10.5281/zenodo.5768332

[4] Federal Emergency Management Agency. (July 29, 2023). *Mold: Problems and Solutions.* https://www.fema.gov/fact-sheet/mold-problems-and-solutions.

rescue efforts and is estimated to have resulted in a net loss of EUR 6.5 billion in residential buildings and household contents[5], with insurers reporting over 400 “major losses”, or those equating to over EUR 1 million in total losses to a single policy holder[6].

Perhaps more concerning, however, is the potential impact on the local affordable housing stock and thus the instability and uncertainty of affordable housing following a flooding disaster. What happened in Houston, Texas should be a cautionary anecdote; in 2017, Hurricane Harvey, a category 4 storm, brought catastrophic flooding. Prior to 2017, Houston, like Ithaca, was already experiencing a housing crisis, resulting in a severe shortage of affordable, accessible rental homes, particularly those available to the lowest income people [7], “80% of all housing stock in Port Arthur and Beaumont was damaged, leaving few options for displaced, low-income families”[8]. Further, when publicly supported, affordable housing is damaged, funds to repair damages are typically stretched thin, as all emergency and rebuilding funds are typically sourced from HUD and FEMA. This reality leaves low-income renters, and particularly renters of color, disproportionately impacted by the effects of a flooding event. Cumulative research has shown that not only are renters disproportionately located in neighborhoods that experience disinvestment, have neglected infrastructure and are therefore more vulnerable to climate-change but, renters are also disproportionately living in older, sub-standard, poorly maintained buildings that are not able to withstand disasters. In fact, from 2015-2017, hurricanes, flooding, and wildfires, among other disasters, damaged more than a half-million rental units, displacing 325,000 renters[9].

Most notably, disasters compound an already complex housing system, underscoring the importance of due consideration of cascading risks. Limited housing stock, exclusionary zoning that limits housing supply, numerous discrete recovery assistance programs with varying timelines, and restrictions on qualifications for support all coalesce to slow community recovery efforts and restrict post-disaster housing to those with ample financial resources[10]. Therefore, communities like Ithaca should continue to focus on inclusive planning practices, including in the upcoming zoning rewrite, to break down historic barriers in access, equity, and inclusionary housing. Though flooding is often

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[5] Verscicherungs. Money Focus (27 August 2021). Flood: GDV significantly raises loss forecast. https://versicherungsprofi.online/branche/assekuranz/flut-gdv-korrigiert-schadenprognose-deutlich-nach-oben_01399/.

[6] Fiedler, Michael. Versicherungsbote. (15 September 2021). GDV: July flood caused more major losses than ever before. https://www.versicherungsbote.de/id/4903294/GDV-Juli-Flut-verursachte-so-viele-Grossschaden-wie-noch-nie/.

[7] Aurand, A., Emmanuel, D, Errico, E. Yentel, D. (March 2017). *The Gap: A shortage of Affordable Homes*. National Low Income Housing Coalition.

[8] Mickelson, Sarah. (21 August 2019). *Impact of Hurricane Harvey: In 2017, Hurricane Harvey Brought Damaging Winds and Flooding to Southeast Texas. The Recovery is Still Ongoing.* Disaster Housing Recovery Coalition.

[9] Office of Policy Development and Research. (2022). The Role of Housing in Climate Change Mitigation and Adaptation. *Evidence Matters, Summer 22.*

[10] Federal Emergency Management Agency; US Department of Housing and Urban Development. (July 2024). Pre-Disaster Housing Planning Initiative. *US Department of Homeland Security*.

thought to be the most imminent climate threat in Ithaca, scorching heat waves are far more likely and pervasive.

Heat

Known as the "silent killer" by labor and climate activists and professionals alike [11, 12], periods of extreme heat can not only cause acute heat-related illness but can also exacerbate existing health conditions. Sub-standard housing conditions, poor ventilation, and the relative scarcity of residential indoor cooling compound the public health risk faced by rising global temperatures. Elderly residents and newborns are particularly vulnerable, especially without adequate access to cooling. For instance, between the years 2000 and 2019, there were 6,364 emergency department visits and/or hospitalizations due heat-related illness and exposure *at home* in New York State; 39% of those cases treated were over the age 75 years of age[13]. These statistics do not include emergency treatment that may have been received after heat exposure in the workplace, during sport or recreation activities, in a public building or institutional home, or other locations – these statistics will be discussed in later sections.

Contributing to the excessive heat risk in Ithaca, is the quality of housing, particularly rental housing, in Upstate New York. In part, the increased risk of tenants is due to the financial mismatch often referred to as the "split incentive" -- a situation where a property owner is not financially motivated to invest in energy upgrades in a rental property, often due to a lack of financial return on the investment or direct benefit to the property owner. This results in housing units that are poorly insulated and lack the essential HVAC upgrades to provide cooling to support the necessary quality of life for tenants. Ruthy Gourevitch, Housing Policy Manager at the Climate and Community Institute, a climate and economy think tank focused on policy development and implementation further describes that the split incentive often leads property owners to seek rent increases:

> *"Landlords who have acquired a rental asset with a goal of generating a good return on their investment are trying to cut back on expenses as much as possible in order to be able to churn a profit. Cutting back on expenses means deferring maintenance a lot over time, leading to very bad conditions for tenants. If a landlord does an upgrade {like energy-efficient HVAC systems}, often their incentive is to increase the value of the property so you can attract higher-earning tenants."*[14]

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[11] Powder, Jackie. (8 July 2024). *Extreme Heat Hazards*. Hopkins Bloomberg Public Health Magazine.
[12] Azzi, Manal; et al. (25 July 2024). Heat at work: Implications for safety and health. International Labor Organization. 75.
[13] Office of Quality and Patient Safety, Division of Information and Statistics. (February 2025). State and region level heat stress hospitalizations and ED visits. New York State Department of Health. *http://www.health.ny.gov/statistics/sparcs/.*
[14] Arnoff, kate. (22 June 2024). When Are We Going to Protect Renters From Extreme Heat? *The New Republic*.

When coupled with climate change, lack of action on housing has the potential to further entrench socio-spatial segregation, housing discrimination, and housing exclusion[15]. By raising the cost of rent in units that have received energy efficiency upgrades, like insulation and cooling, this results in racial and class divides and contributes to gentrification. Further, without a significant investment in building envelopes, energy burden, or the percentage of monthly income spent on energy expenses, will continue to rise, further contributing to the unaffordability of housing.

Real Estate

While the impact of climate change on housing occupants is notable and of primary concern, the economic impacts cannot be ignored. Recent research has shown a national pattern in the real estate market commonly referred to as the "climate housing bubble", whereby properties prone to climate impacts, particularly in flood prone areas, are drastically overvalued[16]. 2023 Researchers found that, nationally, when using the preferred 3% discount rate, properties in flood prone areas were overvalued by $187 billion and that overvaluation was most common in counties without flood disclosure laws. In fact, on average, properties in a 100-year flood zone were overvalued by approximately 8.5%[16]. Please see the figure below for overvaluations by county.

These unaccounted-for flood risks can be attributed to a variety of factors, some of which can be solved by generalized public education. However, some are more systemically entrenched in state and federal systems and will require strategic municipal advocacy or seeking alternative service providers. Namely, experts have drawn critiques of FEMA's Flood Insurance Rate Map (FIRM) development methodology and the stagnant nature of FEMA's flood modeling. For example, FIRMs rely on historical data and are a snapshot of a community's flood risk today. However, they do not take into account changing conditions or how flood risk might change years or decades into the future due to increased or decreased precipitation patterns, fluctuating temperatures, or other factors we know are likely in New York under climate change[17].

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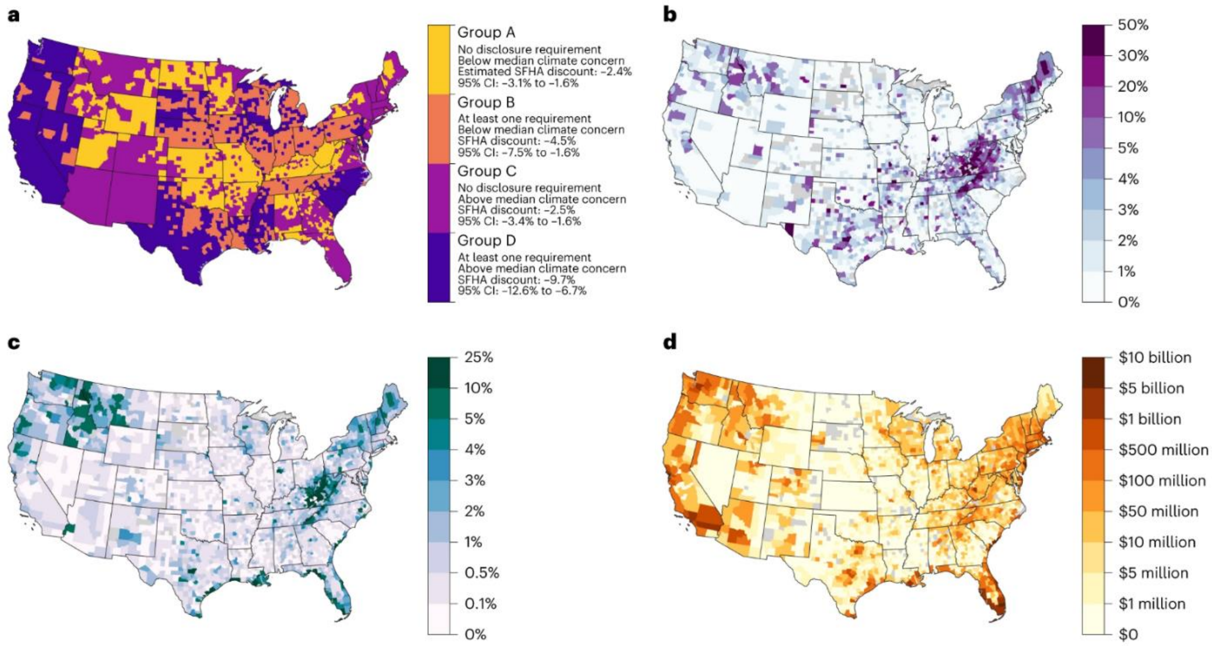
[15] Rajagopal, Balakrishnan. (2023). Towards a just transformation: climate crisis and the right to housing. Report of the SPecial Rapporteur on adequate housing as a component of the right to an adequate standard of living, and on the right to non-discrimination in this context. *United Nations General Assembly, Human Rights Council.* A/HRC/52/28.

[16] Gourevitch, Jesse, et al. (February 2023). Unpriced climate risk and the potential consequences of overvaluation in the US housing markets. *Nature Climate Change*. 13, 250-257. https://doi.org/10.1038/s41558-023-01594-8

[17] Kousky, Carolyn. (2018). Financing Flood Losses: A Discussion of the National Flood Insurance Program. *Risk Management and Insurance Review*. Vol 21, No. 1, 11-32. https://doi.org/10.1111/rmir.12090

From: [Unpriced climate risk and the potential consequences of overvaluation in US housing markets]



a, Estimated flood zone property price discount, differentiated by flood risk disclosure laws and attitudes towards climate change. **b**, Median property-level overvaluation, as a proportion of properties' current fair market value. **c**, Property overvaluation as a proportion of the total fair market value of all properties. **d**, Total overvaluation in dollars. The counties shapefile was downloaded from the US Census Bureau[45].

Labor

Historically, the national Environmental Justice (EJ) movement has deep roots in the labor movement and strong solidarity with unionized workers. In the 1960s, in parallel with the civil rights movement, the American EJ movement was led by communities of color who sought to address the disproportionate environmental burden and lack of protections in their communities. Many scholars, including Dr. Robert Bullard, widely considered the "Father of Environmental Justice"[18], trace the catalyst of this movement back to 1968 when sanitation workers in Memphis, TN were crushed to death by a malfunctioning collection truck[19]. Supported by Dr. Martin Luther King, Jr., the accident and the 1,300-person strike that followed drew nationwide attention to the relationship between pollution, waste, labor, and communities of color.

This deep history between labor and climate concerns was critical to the growth of the environmental and civil rights movements and it is imperative that as climate practitioners and advocates we continue to view these issues as deeply intertwined. Today, the labor and economic market in the Southern Tier and Finger Lakes regions are heavily reliant on

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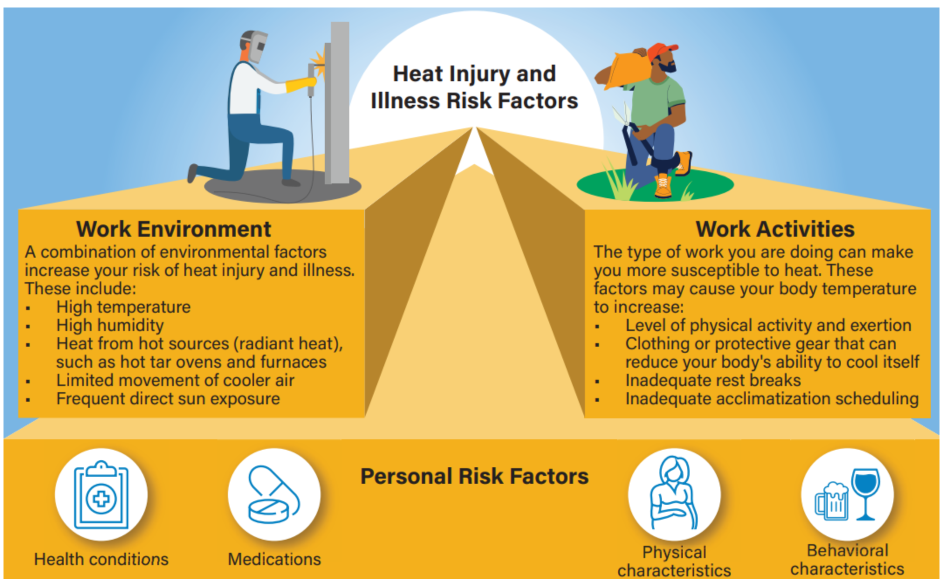
[18] Funes, Yessenia. (19 September 2023). The Father of Environmental Justice Exposes the Geography of Inequity. *Scientific American*.

[19] Stanford University. *Memphis Sanitation Workers' Strike.* The Martin Luther King, Jr. Research and Education Institute.

climatic conditions and our ability to adapt to their variability. Without taking these vulnerabilities into account, we risk susceptibility to destabilization of public health and steady wages, as well as missed opportunities for economic growth.

Heat & Air Quality

Millions of American workers are exposed to dangerous heat levels in their workplaces annually, often leading to cases of illness, reduced productivity, or even fatality. In fact, outdoor workers face a 35-times higher risk of fatality from heat exposure than the general public[20]. Hazardous heat exposure doesn't just occur outdoors, however; it can occur anywhere and during any season if the conditions are right, not just during heat waves[21]. It's important to note that some workers are at an increased risk of experiencing heat illness or heat stress, particularly those with pre-existing conditions, the elderly, pregnant people, and those who are not properly acclimated to higher temperatures. Other factors contribute to a worker's relative risk[22] as well; see the graphic below and its associated footnote for more information.



The dangers of extreme heat are two-fold, presenting both public health and economic challenges. Details of how heat illness can affect an individual's health will be discussed in greater depth in the *Public Health* section. According to climate data, 2024 was the hottest

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[20] Gubernot, Diane M., G. Brooke Anderson, and Katherine L. Hunting. 2015. "Characterizing Occupational Heat-Related Mortality in the United States, 2000–2010: An Analysis Using the Census of Fatal Occupational Injuries Database." American Journal of Industrial Medicine 58 (2): 203–211. https://doi.org/10.1002/ajim.22381.

[21] *Working in Outdoor and Indoor Heat Environments.* (January 2021). Occupational Safety and Health Administration. https://www.osha.gov/heat-exposure

[22] *Personal Risk Factors and Heat Exposure*. (2023). Occupational Safety and Health Administration. https://www.osha.gov/sites/default/files/publications/OSHA4374.pdf

year on record in seventeen states, including New York[23]. With rising temperatures, the number of hours and days when outdoor work is considerably unsafe will continue to increase, forcing workers to choose between their health and a paycheck[24].

In Tompkins County, 14% of workers aged 16 years and older work outdoors, or a total of 7,194 individuals representing over $202 million in annual earnings[25]. The majority of these workers are in the building and grounds maintenance; construction and extraction; installation, maintenance, and repair; and transportation sectors, though protective services also represents a large number of workers. Under the RCP8.5, or low action/high emissions, scenario, climate models predict a total of 8 extreme heat days per year in Tompkins County by late century; in contrast, in the most unlikely scenario where rapid action is taken to mitigate climate change and global average warming is capped at 2°C, extreme heat days are limited to 1 per year[26]. By normalizing the RCIP8.5 trend across all labor sectors and all workers at risk for lost wages, the Union of Concerned Scientists estimates that annually, $6,658,508 will be lost in collective yearly worker pay[26].

When these financial impacts are extrapolated over time, we can see the effects beyond daily wages and begin to appreciate the risk associated with the number of increased heat waves on the local labor market. If heat waves last longer in a given year and create operational conditions that necessitate the closure of businesses, workers could face furlough or permanently reduced hours. When this becomes a pattern, prolonged time away may cause workers to lose employer-provided benefits, like health insurance, paid leave, or employee assistance, exacerbating financial and health impacts[26].

Despite the risk of lost wages, job loss, and negative health effects, workers most vulnerable to these impacts have not historically received an increase in wages to compensate for these increased risks.

Further, because Black and Latinx workers are disproportionately represented in occupations frequently exposed to the outdoors and extreme heat, the potential for lost wages only exacerbates inequity in poverty rates and economic mobility, all of which result from centuries of systemic racism[25]. The unjust exposure and burden of the impacts of

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[23] *2024 was nation's warmest year on record.* National Oceanic and Atmospheric Administration. (10 January 2025). https://www.noaa.gov/news/2024-was-nations-warmest-year-on-record
[24] Dahl, Kristina and Licker, Rachel. 2021. *Too Hot to Work: Assessing the Threats Climate Change Poses to Outdoor Workers.* Cambridge, MA: Union of Concerned Scientists. https://doi.org/10.47923/2021.14236
[25] Union of Concerned Scientists. 2021. *UCS Too Hot To Work Data.* https://www.essoar.org/doi/abs/10.1002/essoar.10507713.1
[26] "Impacts of Climate Change on American Household Finances" (September 29, 2023). *U.S. Department of Treasury.*

climate change on communities of color will be discussed in more detail in the *Racial Equity* section.

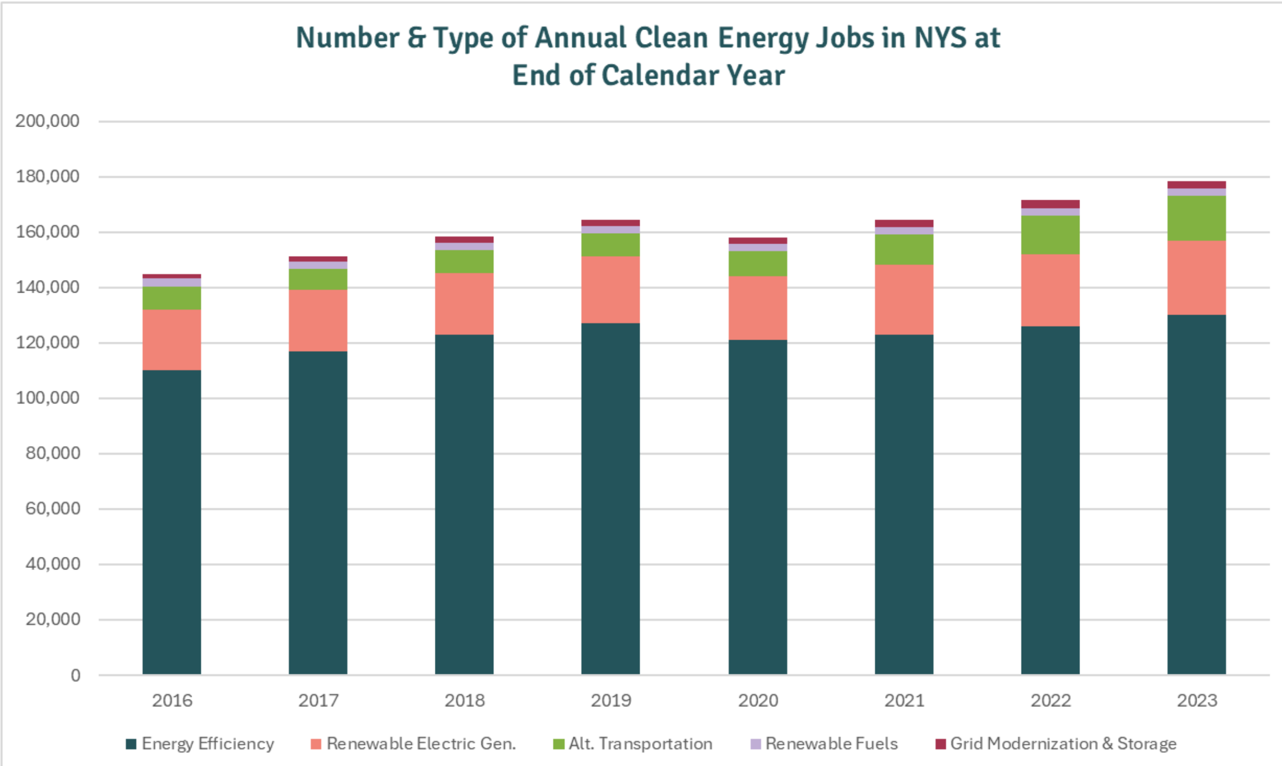
Market Change

New York State is at the forefront of climate action in the United States, which was underscored by the adoption of the Climate Leadership and Community Protection Act (CLCPA) in 2019. The CLCPA sets ambitious statewide targets for reducing greenhouse gas emissions, including achieving an 85% decrease by 2050. However, labor growth has signaled that New York State has been heading in the direction of clean, green energy since at least 2016, when the state energy agency, NYSERDA, began producing annual reports. The steady growth, and strong recovery following the COVID-19 pandemic, of clean energy jobs emphasizes the stability and potential trajectory in the green jobs sector and, unless the City of Ithaca and surrounding areas are poised to take advantage of these market opportunities, the potential opportunity loss for career-seeking residents and host-community-seeking businesses.

The New York State Energy Research and Development Authority, or NYSERDA, has been collecting and reporting on clean energy job growth across building, transportation, generation, and storage sectors since 2016. With the exception of the years spanning the height of the COVID-19 pandemic, New York has experienced record clean energy job growth. In fact, even at the peak of the pandemic, New York State lost only 9.6% of its total clean energy workforce compared to the nationwide average of 14%[27]. Since 2016, New

[27] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2020.* (NYSERDA, 2021).

Yorkers have seen a net 22% increase in clean energy jobs across the state, with growth remaining steady since the pandemic[28,29,30,31,28,32,33,34].



More recently, clean energy jobs outpaced growth in any other economic sector by more than double in New York State, while also boasting entry-level positions earning 12% more

[28] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2016*. (NYSERDA, 2017).

[29] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2017*. (NYSERDA, 2018).

[30] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2018*. (NYSERDA, 2019).

[31] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2019*. (NYSERDA, 2020).

[32] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2021*. (NYSERDA, 2022).

[33] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2022*. (NYSERDA, 2023).

[34] New York State Energy Research & Development Authority. *New York Clean Energy Industry Report 2023*. (NYSERDA, 2024).

than the same job in comparable industries and 93% of workers anticipating career advancement in the next 12 months[35], indicating significant opportunity for long-term careers for Ithaca jobseekers. The rapid job growth NYS is experiencing is partly due to national deployment of clean energy technologies, and the projected deployment into 2030. The National Renewable Energy Laboratory (NREL) projects between 35,370 (Regional Energy Deployment Systems (ReEDS) mid-case scenario or "business as usual") and 51,822 (accelerated technology deployment scenario) New York jobs in the solar photovoltaics, land-based wind energy, grid-connected battery storage, and building energy efficiency sectors will be needed to reach 2030 projections[35].

While impressive on their own, NREL projections do not include the number or type of jobs required to meet NYS Climate Leadership and Community Protection Act. In 2023, the Just Transition Working Group (JTWG), a team part of the NYS Climate Action Council, produced an update to their 2021 Jobs Study, which provided employment and workforce analyses, economic impact models, and employment forecasts. Based on these analyses, the JTWG projects each of the five regions of NYS will see an increase of at least 10,000 net new "green" jobs by 2030[36]. Further, the City of Ithaca and the Cornell University Public Interest Technology Initiative (PiTech) produced our own modeling to determine the number of additional HVAC installers, plumbers, and electricians needed in the city workforce to meet Ithaca Green New Deal residential building decarbonization goals. The statistical model showed that, at minimum, 25 HVAC installers, 18 plumbers, and 8 electricians need to be added to the workforce quarterly between January 2024 and the end of 2030 [37].

Extensive research and modeling shows that there is ample opportunity for economic and labor growth in the clean energy sector. However, unless Ithaca is actively training, upskilling, and retaining clean energy workers, the potential for this growth will be lost to other surrounding areas. This concern for retention of local clean energy talent is exacerbated when local workforce training programs do not match the goals of municipal economic development goals and strategies. To achieve sustainable economic growth and inclusive prosperity, it is recommended that local governments and community-based

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[35] National Renewable Energy Laboratory. *New York's Clean Energy Job's Potential Through 2030.* U.S. Department of Energy (March 2022).
[36] Just Transition Working Group. *2021 Jobs Study, March 2023: Vintage Update*. BW Research Partnership (March 2023).
[37] Cho, Jason. *Workforce Modelling for 2030 Electrification Goal.* Cornell University Pitech Siegel PhD Fellowship, Cornell University (August 2023).

organizationsmarket analyses to ensure local employability within the municipality of residence is possible [38].

Migrant Rights & Job Access

The City of Ithaca has repeatedly affirmed its commitment to being "a sanctuary for all hard-working people" [39]. The original resolution proclaiming Ithaca's sanctuary status was adopted in 1985 during the Guatemalan and Salvadoran refugee crisis [40], and was strengthened and reaffirmed in 2017 [41] and again in 2025 [42] in response to Presidential threats of migrant deportation and incidents of reported violence. Ithaca's sanctuary status, coupled with a warm, welcoming culture, temperate climate, and insulation from coastlines could make it a desirable area for migrant relocation. While concern for an influx migrants has grown recently, numerous factor make this an unlikely reality. However, given the local concern, the economic risks will be described briefly below.

Slow-onset climate change is considered a major driver of global migration. Within this context, climate interacts with other variables, including lack of decent work, weak governance, and intercommunity violence, forcing people to relocate domestically and internationally [43]. Unfortunately, in most communities, economic development workforce planning does not account for the potential influx of migrants, or the potential skillsets and experiences they may bring with them. This presents the City of Ithaca and surrounding communities with a conundrum: Proactively plan and prepare for the potential increased population with varying skills, or lean into the economic sectors that have historically proven successful in Ithaca, limiting job-*type* variability in the future but safeguarding against risk of too few resources for too many people. A lack of employment opportunities compounds livability in other sectors, particularly in an unstable or unaffordable housing market, stressing social services and diminishing public health and safety [44].

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[38] Better Buildings Workforce Accelerator. *Program Design and Evaluation.* Better Buildings: U.S. Department of Energy (2022).
[39] Myrick, S. "This is an ordinance — not just a statement — but an actual ordinance that will ensure Ithaca remains a sanctuary for all hard-working people," Facebook, February 1, 2017.
[40] Merina, V. *Cities vs. The INS: Reviving an Old Concept.* Los Angeles Times (1985).
[41] VI. Ithaca, NY, *Sanctuary City,* (City of Ithaca, 2017), sec. 215-37 – 215-46. https://ecode360.com/32288260
[42] City of Ithaca, NY. Common Council. *A Resolution Reaffirming the City of Ithaca's Commitment to Human Rights Protection for Migrants, Reproductive Rights, and Gender Affirming Care* (5 February 2025). *https://d2kbkoa27fdvtw.cloudfront.net/cityofithaca/e3680b43a22698ee37f6b1b288dabbe60.pdf*
[43] International Labor Organization. *Human mobility, climate change and a just transition.* United Nations. https://www.ilo.org/migration-stub-9231/human-mobility-climate-change-and-just-transition
[44] Chishti, M., Putzel-Kavanaugh, C. *After Crisis of Unprecedented Migrant Arrival, U.S. Cities Settle into New Normal* (1 August 2024). Migration Policy Institute.

Migrant success in a new city is also highly dependent on where they're migrating from, their spoken language, and whether a work permit is required to gain employment. Without fundamental language and writing skills, individuals may struggle to find permanent employment in high-skilled jobs, despite extensive education and training. This is largely due to the difficulties and wait times associated with receiving a U.S. work authorization, which average three months to process [45]. Coupled with limited English language proficiency and/or low Test of English as a Foreign Language (TOEFL) scores, this creates a nearly impossible ecosystem for migrants reentering the workforce. Minnesota migrants experienced this first-hand, when physicians and pharmacists, despite exceptional need in the healthcare sector nationwide, found themselves working at coffee shops or jobs where their pay was one-third the market norm for their degree(s) [45].

Risk in the labor sector, like all others, only compounds or, in some cases, multiplies risk in other sectors. Lack of employment creates affordable housing pressure, strains social services, and weakens government's ability to respond to disaster. While the current housing and job market, coupled with transportation challenges, do not lead staff to believe migration will become a critical issue within the City, the risk increases exponentially with state or federal immigration policy intervention, potentially leading to higher levels of government determining where migrants end up settling.

Public Health

The City of Ithaca's public health landscape is increasingly vulnerable to the impacts of climate change. Main threats include the rising risks of air pollution, extreme temperatures, vector-borne diseases, and the stress on water and food systems. The following section outlines the specific risks posed by these climate-related threats and their potential effects on public health in Ithaca.

Outdoor Health

Air Toxins (PM 2.5, PM 10, and Other Pollutants)

Fine PM 2.5 and coarse PM 10 particulate matter, tiny particles that can be inhaled deep into the lungs[46], are on the rise across the Northeast the due to more frequent regional

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[45] Delion, N. *Lost wages, opportunities: College-educated immigrants find many barriers to high-skilled jobs in the U.S.* (19 March 2024). Sahan Journal.

[46] Environmental Protection Agency. (2024a, June 20). *Particulate Matter (PM) Basics*. EPA. https://www.epa.gov/pm-pollution/particulate-matter-pm-basics

wildfires[47] and an increased reliance on fossil fuel combustion during extreme temperatures[48] (EIA, 2024; Blackmon, 2022; U.S. Department of Energy, 2018). These particles are often invisible but can have profound effects on human health, particularly for children, seniors, outdoor workers, and individuals with preexisting heart or respiratory conditions[49] such as emphysema, asthma, chronic bronchitis, and COPD ("Air Quality", n.d.).

Events like the 2023 Canadian wildfire season, which blanketed Ithaca and much of the Northeast in smoke for days, demonstrate how rapidly air quality can deteriorate[50]. On June 6, 2023, Ithaca recorded an Air Quality Index (AQI) of 162, classified as "unhealthy", due to drifting wildfire smoke[51]. Beyond acute events, chronic exposure to elevated PM 2.5 levels, often caused by residential burning (trash, leaves, and brush burning, bonfires, etc.), vehicle emissions, and industrial pollutants, poses long-term health risks[52], with health costs exceeding $800 billion per year[53]. Sustained PM 2.5 exposure has been linked

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[47] Kerr, G. H., DeGaetano, A. T., Stoof, C. R., & Ward, D. (2018). Climate change effects on wildland fire risk in the Northeastern and Great Lakes states predicted by a downscaled multi-model ensemble. *Theoretical and Applied Climatology*, *131*(1), 625–639. https://doi.org/10.1007/s00704-016-1994-4

[48] Blackmon, D. (2022, December 27). *ISO New England pumps up the fuel oil again during Winter Storm*. Forbes. https://www.forbes.com/sites/davidblackmon/2022/12/26/iso-new-england-pumps-up-the-fuel-oil-again-during-winter-storm/?sh=2183b30b1482

[49] Environmental Protection Agency. (2024b, July 16). *Health and Environmental Effects of Particulate Matter (PM)*. EPA. https://www.epa.gov/pm-pollution/health-and-environmental-effects-particulate-matter-pm

[50] Yu, M., Zhang, S., Ning, H., Li, Z., & Zhang, K. (2024). Assessing the 2023 Canadian wildfire smoke impact in Northeastern US: Air quality, exposure and environmental justice. *Science of The Total Environment*, *926*, 171853. https://doi.org/10.1016/j.scitotenv.2024.171853

[51] Butler, M. (2023, June 7). *Update: Local Air Quality Now "unhealthy," Health Department encourages staying indoors*. The Ithaca Voice. https://ithacavoice.org/2023/06/air-quality-alert-issued-locally-until-tuesday-due-to-wildfires-in-canada/

[52] Environmental Protection Agency. (2024a, June 20). *Particulate Matter (PM) Basics*. EPA. https://www.epa.gov/pm-pollution/particulate-matter-pm-basics

[53] *The Costs of Inaction: The Economic Burden of Fossil Fuels and Climate Change on Health in the United States.* (2021). NRDC and The Medical Society Consortium on Climate and Health. https://www.nrdc.org/sites/default/files/costs-inaction-burden-health-report.pdf

to increased rates of heart attacks, asthma, respiratory issues, and irregular heartbeat[54]. In the United States, air pollution is linked to around 100,000 premature deaths each year[55]. In neighborhoods where residents may live near busy roadways[56], such as near Route 13 and Downtown Ithaca, or lack access to air conditioning or filtration[57], the effects are even more pronounced.

Heat (Extreme Temperatures)

Nationally, heat is the leading preventable weather-related cause of death, killing more people annually than hurricanes, floods, and tornadoes combined[58]. Each year, 750 to 1,300 Americans die from extreme heat, and that number is expected to rise as climate change accelerates[59]. Historically known for its temperate summers[60], Ithaca is now facing rising temperatures and more frequent extreme heat days due to climate change[61].

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[54] Environmental Protection Agency. (2024b, July 16). *Health and Environmental Effects of Particulate Matter (PM)*. EPA. https://www.epa.gov/pm-pollution/health-and-environmental-effects-particulate-matter-pm

[55] Thakrar, S. K., Balasubramanian, S., Adams, P. J., Azevedo, I. M. L., Muller, N. Z., Pandis, S. N., Polasky, S., Pope, C. A. I., Robinson, A. L., Apte, J. S., Tessum, C. W., Marshall, J. D., & Hill, J. D. (2020). Reducing Mortality from Air Pollution in the United States by Targeting Specific Emission Sources. *Environmental Science & Technology Letters*, *7*(9), 639–645. https://doi.org/10.1021/acs.estlett.0c00424

[56] Wu, C. D., MacNaughton, P., Melly, S., Lane, K., Adamkiewicz, G., Durant, J. L., Brugge, D., & Spengler, J. D. (2014). Mapping the vertical distribution of population and particulate air pollution in a near-highway urban neighborhood: implications for exposure assessment. *Journal of exposure science & environmental epidemiology*, *24*(3), 297 304.https://doi.org/10.1038/jes.2013.6

[57] Chuang, H.-C., Ho, K.-F., Lin, L.-Y., Chang, T.-Y., Hong, G.-B., Ma, C.-M., Liu, I.-J., & Chuang, K.-J. (2017). Long-term indoor air conditioner filtration and cardiovascular health: A randomized crossover intervention study. *Environment International*, *106*, 91–96. https://doi.org/ 10.1016/j.envint.2017.06.008

[58] National Oceanic and Atmospheric Administration. (n.d.). *Weather related fatality and injury statistics*. National Weather Service. https://www.weather.gov/hazstat/

[59] Environmental Protection Agency. *Climate Change Indicators: Heat-Related Deaths*. EPA. (2025, February 26). https://www.epa.gov/climate-indicators/climate-change-indicators-heat-related-deaths

[60] *The Ithaca Climate Page*. Northeast Regional Climate Center. (n.d.a). https://www.nrcc.cornell.edu/wxstation/ithaca/ithaca.html

[61] *Climate Change in New York State.* (2014, September). New York State Energy Research and Development Authority https://www.nyserda.ny.gov/-

Between 1970 and 2020, the average number of days above 90°F in the region nearly doubled, and projections show that Ithaca could experience up to 20 such days per year by mid-century[62]. In 2023, Ithaca had an average temperature of 48.3°F[63] , 2.0°F above the 1991-2020 annual average of 46.3°F[64].

This marks a large shift for a city where many homes and public buildings were not built with air conditioning or passive cooling in mind. Even homes with central air conditioning or that utilize portable cooling units, the rising cost of utilities over the past three years has increased the summer energy burden beyond what some residents can comfortably afford. More information on energy burden and its impact on access to heating and cooling will be discussed in the *Power Reliability* section.

Extreme heat poses a serious public health threat, particularly for infants, seniors, individuals with chronic illnesses, and outdoor workers[65]. Heat stress, which results from the body's inability to rid itself of internally generated heat, is the leading cause of weather-related deaths and can lead to heat related illnesses such as heatstroke[66]. Urban heat islands, areas with more pavement and less tree cover, lead to higher temperatures which worsen these effects, especially in densely populated or lower-income neighborhoods[67]. Residents without access to air conditioning, green space, or public cooling centers face

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/media/Project/Nyserda/Files/Publications/Research/Environmental/ClimAID/2014-ClimAid-Report.pdf

[62] Lamie, C., Bader, D., Graziano, K., Horton, R., John, K., O'Hern, N., Spungin, S., & Stevens, A. (2024). New York State Climate Impacts Assessment Chapter 02: New York State's Changing Climate. Ann NY Acad Sci., 1542, 91–145. https://doi.org/10.1111/nyas.15240

[63] *The Ithaca Climate Page*. Northeast Regional Climate Center. (n.d.b). https://www.nrcc.cornell.edu/wxstation/ithaca/normal.html

[64] *The Ithaca Climate Page*. Northeast Regional Climate Center. (n.d.b). https://www.nrcc.cornell.edu/wxstation/ithaca/normal.html

[65] *Who is at risk to extreme heat*. National Integrated Heat Health Information System. (n.d.-a). https://www.heat.gov/pages/who-is-at-risk-to-extreme-heat

[66] *Heat and health*. (2024, May 21). World Health Organization. https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health

[67] Walker, J. (2024, August 13). *Urban heat islands and a climate of inequities*. University of Michigan School of Environment and Sustainability. https://seas.umich.edu/news/urban-heat-islands-and- climate-inequities-0

higher exposure and greater risk of heat-related illness or death[68]. In addition to rising overall temperatures, rapid weather swings continue to have severe impacts. For example, in September 2020, the Rocky Mountains experienced a sudden shift from a severe heatwave to heavy snowfall, with a temperature drop of over 68°F (20°C) within a day, which led to power outages and property damage[69]. Similarly, in April 2021, much of Europe faced an abrupt transition from warm to cold weather that caused widespread crop frost damage[70]. This abrupt transition from warm to cold weather could similarly impact Tompkins County by increasing the risk of sudden spring frosts that damage budding crops and threaten local agricultural yields.

Pests and Vector -Borne Diseases

As warmer temperatures become more common across New York State, the rates of vector-borne diseases, diseases transmitted from animals to humans, are rising. Climate change has increased temperatures in New York State, resulting in longer warm seasons and milder winters[71]. These increased temperatures have increased the activity of young ticks, which are known to be most active between mid-May to mid-August, in contrast to adult ticks that are more likely to carry disease, which are most active from March to mid-

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[68] Mann, R., & Schuetz, J. (2022, July 25). *As extreme heat grips the globe, access to air conditioning is an urgent public health issue*. The Brookings Institution. https://www.brookings.edu/articles/as-extreme-heat-grips-the-globe-access-to-air-conditioning-is-an-urgent-public-health-issue/#:~:text=Longer%2Dterm%20options%20include%20expanding,more%20climate%2Dfriendly%20land%20use.

[69] Wu, C. D., MacNaughton, P., Melly, S., Lane, K., Adamkiewicz, G., Durant, J. L., Brugge, D., & Spengler, J. D. (2014). Mapping the vertical distribution of population and particulate air pollution in a near-highway urban neighborhood: implications for exposure assessment. *Journal of exposure science & environmental epidemiology*, *24*(3), 297 304.https://doi.org/10.1038/jes.2013.6

[70] Wu, C. D., MacNaughton, P., Melly, S., Lane, K., Adamkiewicz, G., Durant, J. L., Brugge, D., & Spengler, J. D. (2014). Mapping the vertical distribution of population and particulate air pollution in a near-highway urban neighborhood: implications for exposure assessment. *Journal of exposure science & environmental epidemiology*, *24*(3), 297 304.https://doi.org/10.1038/jes.2013.6

[71] Lin, S., Shrestha, S., Prusinski, M. A., White, J. L., Lukacik, G., Smith, M., Lu, J., & Backenson, B. (2019). The effects of multiyear and seasonal weather factors on incidence of Lyme disease and its vector in New York State. *The Science of the total environment*, *665*, 1182–1188. https://doi.org/10.1016/j.scitotenv.2019.02.123 ;

Climate change and health. New York State Department of Health . (2024, December). https://www.health.ny.gov/environmental/weather/

May and again from mid-August to November during warmer weather[72]. The most common tick-borne disease in New York State, Lyme disease, is caused by a bacteria passed to humans through the bite of infected blacklegged ticks[73] . Symptoms often include fever, headache, fatigue, facial paralysis, and a skin rash called erythema migrans[74]. The Centers for Disease Control and Prevention (CDC) reports that incidence rates of Lyme Disease in New York State have increased from 4,615 cases in 2013 to 22,173 cases in 2023 (CDC, 2025). In Tompkins County, the number of reported Lyme disease cases nearly doubled, increasing from 265 in 2022 to 535 by the end of August 2023[75].

Climate impacts are causing a rise in additional tick-borne diseases, including anaplasmosis and babesiosis, both of which present with flu-like symptoms[76]. Other vector-borne diseases such as West Nile Virus and Eastern Equine Encephalitis, which are transmitted by mosquitoes, are also increasing in New York State. In 2024, 100 human cases of West Nile Virus were reported outside of New York City[77] compared to 64 in 2023[78]. In 2024, New York State also saw its first case of Eastern Equine Encephalitis since

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[72] *Be Tick Free- A Guide for Preventing Lyme Disease.* New York State Department of Health. (2023, July). https://www.health.ny.gov/publications/2825/

[73] *About Lyme Disease.* (2024, August 26). The Centers for Disease Control and Prevention. https://www.cdc.gov/lyme/about/index.html#:~:text=called%20erythema%20migrans.-, If%20left%20untreated%2C%20infection%20can%20spread%20to%20joi nts%2C%20the%20heart,performed%20with%20FDA%2Dcleared%20tests

[74] *Lyme Disease Rashes*. (2024, May 15). Centers for Disease Control. https://www.cdc.gov/lyme/signs-symptoms/lyme-disease-rashes.html

[75] Jordan, J. *Tick-borne illnesses jump in Tompkins County.* (2023, September 15). Ithaca Voice. https://ithacavoice.org/2023/09/tick-borne-illnesses-jump-in-tompkins-county/

[76] *About Anaplasmosis*. (2024, September 4). Centers for Disease Control. https://www.cdc.gov/anaplasmosis/about/index.html#:~:text=Anaplasmosis%20is%20a%2 0disease%20caused,%2C%20chills%2C%20and%20muscle%20aches;

About Babesiosis. (2024, February 12). Centers for Disease Control. https://www.cdc.gov/babesiosis/about/index.html#:~:text=Babe siosis%20is%20a%20disease%20caused,others%20have%20flu%2Dlike%20symptoms.

[77] *2024 Mosquito-borne Illness Annual Report*. (2024). New York State Department of Health. https://www.health.ny.gov/diseases/mosquitoes/repo rts/2024/docs/summary_report.pdf

[78] *2023 Mosquito-borne Illness Annual Report*. (2023). New York State Department of Health. https://www.health.ny.gov/diseases/mosquitoes/reports/2023/docs/summary_report.pdf

2015[79]. Although these diseases are not currently widespread in Tompkins County, changing climate conditions could increase their potential risk in the future.

Water Quality and Reliability

Six Mile Creek serves as the primary source of drinking water for the City of Ithaca[80]. Utilizing a gravity-fed system, water flows from a 60-foot reservoir directly to the water treatment plant[81]. This system is supported by a predominantly forested watershed covering 46.4 square miles, which plays a critical role in naturally filtering and maintaining water quality. However, climate change is intensifying risks to both water quality and supply reliability.

Increasingly intense and frequent storm events are likely to generate rapid runoff, which in turn can mobilize excessive sediment and pollutants, including agricultural and urban contaminants, into the creek[82]. This surge in turbidity and contaminant load poses challenges for the overall capacity of the water treatment plant, adding costs and operational stresses to an already delicate system[83]. Contaminants in turn increase concentrations of harmful substances and pathogens in the water, posing health risks if

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[79] *Eastern Equine Encephalitis*. (2024, September). New York State Department of Health. https://www.health.ny.gov/diseases/communicable/eastern_equine_encephalitis/#:~:text=Anyone%20can%20be%20infected%20with,last20cases%20were%20in%202015.

[80] *2024 Annual Drinking Water Quality Report.* (2024). Ithaca Water. https://www.cityofithaca.org/DocumentCenter/View/15429 /AWQR-2023?bidId=

[81] *2024 Annual Drinking Water Quality Report.* (2024). Ithaca Water. https://www.cityofithaca.org/DocumentCenter/View/15429 /AWQR-2023?bidId=

[82] Six Mile Creek: A Status Report. (2007, May). City of Ithaca. https://www.cityofithaca.org/DocumentCenter/View/496/Six-Mile-Creek -Status-Report-May-2007- PDF

[83] Six Mile Creek Source Water Assessment Report. (n.d.) New York State Department of Health. https://www.cityofithaca.org/DocumentCenter/View/10143/Six-Mile-Creek-Source-Water- Assessment-Report;

Six Mile Creek: A Status Report. (2007, May). City of Ithaca. https://www.cityofithaca.org/DocumentCenter/View/496/Six-Mile-Creek -Status-Report-May-2007- PDF consumed[84] . As weather patterns change, prolonged droughts[85] can concurrently reduce base flows in the creek, which may result in higher concentrations of natural and human-caused contaminants[86] and impair the efficiency of the gravity-fed water delivery system. Rising ambient temperatures further exacerbate these problems by promoting the proliferation of pathogens and altering the natural biochemical processes that typically help to purify the water, potentially reducing the effectiveness of existing treatment measures[87]. Rising temperatures can increase the release of lead from aging pipes into drinking water, particularly during heat waves, which are becoming more frequent due to climate change. This raises concerns about potential lead exposure, especially for vulnerable populations such as children[88].

Nutrition & Culturally Appropriate Foods

Climate change poses significant threats to nutrition[89] and the availability of culturally appropriate foods in Ithaca. The local food system is increasingly vulnerable to climate-induced disruptions, which can compromise the nutritional quality of food[90] and access to

[84] Delpla, I., Jung, A.-V., Baures, E., Clement, M., & Thomas, O. (2009). Impacts of climate change on surface water quality in relation to drinking water production. *Environment International*, *35*(8), 1225–1233. https://doi.org/10.1016/j.envint.2009.07.001

[85] Friedlander, B. (2024, February 8). *NYS Agricultural Assessment Cultivates Climate Crisis Solutions*. Cornell Chronicle. https://news.cornell.edu/stories/2024/02/nys-agricultural-assessment-cultivates-climate-crisis-solutions?utm_source=chatgpt.com

[86] *Health impacts of drought*. (2024, March 28). Centers for Disease Control and Prevention. https://www.cdc.gov/drought-health/health-implications/index.html

[87] Levy, K., Smith, S. M., & Carlton, E. J. (2018). Climate Change Impacts on Waterborne Diseases: Moving Toward Designing Interventions. *Current environmental health reports*, *5*(2), 272-282. https://doi.org/10.1007/s40572-018-0199-7

[88] Gopal K. An EPA rule will reduce lead in drinking water-unless this effort to block it succeeds. Inside Climate News. February 13, 2025. Accessed April 29, 2025. https://insideclimatenews.org/news/13022025/epa-rule-to-reduce-drinking-water-lead-potential-reversal/.

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culturally significant dietary options[91]. In Tompkins County, climate change has led to increased drought, erratic weather patterns, and other environmental stressors that challenge local agriculture[92]. These conditions make it more difficult to distribute and maintain fresh produce, leading many small retailers and food pantries to rely on shelf-stable options[93]. Such foods often fail to meet the nutritional and cultural needs of the community, particularly affecting those who depend on emergency food programs[94].

Climate change is also expected to disrupt agriculture and global food supply chains, leading to lower crop yields, higher costs, and reduced access to diverse, nutritious, and affordable foods in places like Tompkins County[95]. This increase in poor nutrition can lead to weakened immune systems, nutrient deficiency disorders, and malnourishment[96].

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[91] Raj, S., Roodbar, S., Brinkley, C., & Wolfe, D. W. (2022). Food Security and Climate Change: Differences in Impacts and Adaptation Strategies for Rural Communities in the Global South and North. *Frontiers in Sustainable Food Systems*, *Volume 5-2021*. https://www.frontiersin.org/journals/sustainable-food-systems/artic les/10.3389/fsufs.2021.691191

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Fulltime Outdoor Living and Exposure

Climate change poses significant risks to individuals in Ithaca who rely on full-time outdoor living, particularly those experiencing homelessness. The city has witnessed a 38% increase in homelessness between 2022 and 2024, with over 100 adults awaiting supportive housing[97]). This growing population faces heightened exposure to climate-related hazards. Projected climate models indicate that Ithaca will experience approximately six to seven heavy rain events and three to six heat waves annually by the 2050s[98]. Such conditions exacerbate the vulnerability of unhoused individuals, who often lack access to shelter and essential services. Additionally, many of Ithaca's homeless encampments are situated in flood-prone areas, increasing the risk of displacement and health hazards during extreme weather events[99].

Indoor Health

Ventilation and Indoor Air Quality

Climate change is poised to continue significantly impacting weather, likely causing substantial increases in average and peak summer temperatures, which in turn will lead to elevated indoor temperatures[100]. Increasing frequency and severity of extreme weather events, such as heat waves, cold snaps, and storms, can disrupt the performance of HVAC systems[101]. Warming temperatures are especially concerning when it comes to indoor

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[97] Cohen, E. (2025, April 16). *After loss of SJCS shelter, Tompkins County shifts toward long-term homelessness strategy*. The Cornell Daily Sun. https://www.cornellsun.com/article/2025/04/after-loss-of-sjcs-shelter-tompkins-county-shifts-toward-long-term-homelessness-strategy

[98] *Climate Change in New York State.* (2014, September). New York State Energy Research and Development Authority https://www.nyserda.ny.gov/-/media/Project/Nyserda/Files/Publications/Research/Environmental/ClimAID/2014-ClimAid-Report.pdf

[99] *Ithaca, NY Flood Map and climate risk report.* (2025). First Street Technology. https://firststreet.org/city/ithaca-ny/3638077_fsid/flood.

[100] Zhao, J., Uhde, E., Salthammer, T., Antretter, F., Shaw, D., Carslaw, N., & Schieweck, A. (2024). Long-term prediction of the effects of climate change on indoor climate and air quality. *Environmental Research*, *243*, 117804. https://doi.org/10.1016/j.envres.2023.117804

[101] Bell, N. O., Bilbao, J. I., Kay, M., & Sproul, A. B. (2022). Future climate scenarios and their impact on heating, ventilation and air-conditioning system design and performance for commercial buildings for 2050. *Renewable and Sustainable Energy Reviews*, *162*, 112363. https://doi.org/10.1016/j.rser.2022.112363

ventilation. Warming can raise indoor relative humidity, which increases the risk of mold growth, particularly on sensitive materials like wood or in areas with thermal bridging. These effects are especially prevalent in older or poorly ventilated buildings[102] . Indoor overheating during summer months may exceed thermal comfort thresholds, posing health risks, especially for vulnerable populations such as the elderly or those with respiratory conditions[103] . As temperatures rise, the emission rates of VOCs like limonene from household products and materials may increase, while ventilation may not sufficiently offset the accumulation of such pollutants[104]. Health effects of VOC exposure can include ear, nose, and throat irritation, dizziness, nausea, and damage to kidneys and the nervous system[105]. Ozone intrusion and indoor chemical reactions could also shift, potentially degrading indoor air quality despite decreasing outdoor ozone levels in some scenarios[106]. Exposure to ozone can cause inflammation and damage of airways and aggravate existing lung conditions[107].

Lack of Cooling

As the frequency and severity of extreme heat events increases, lack of adequate cooling, particularly in residential buildings, public housing, and community facilities, poses a growing public health risk. Prolonged exposure to high indoor temperatures can lead to heat-related illnesses such as heat exhaustion, heat stroke, and the worsening of preexisting conditions like cardiovascular and respiratory diseases[108]. These risks are especially acute for vulnerable populations, including older adults, young children, people

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[102] Zhao, J., Uhde, E., Salthammer, T., Antretter, F., Shaw, D., Carslaw, N., & Schieweck, A. (2024). Long-term prediction of the effects of climate change on indoor climate and air quality. *Environmental Research*, *243*, 117804. https://doi.org/10.1016/j.envres.2023.117804

[103] Zhao, J., Uhde, E., Salthammer, T., Antretter, F., Shaw, D., Carslaw, N., & Schieweck, A. (2024). Long-term prediction of the effects of climate change on indoor climate and air quality. *Environmental Research*, *243*, 117804. https://doi.org/10.1016/j.envres.2023.117804

[104] Zhao, J., Uhde, E., Salthammer, T., Antretter, F., Shaw, D., Carslaw, N., & Schieweck, A. (2024). Long-term prediction of the effects of climate change on indoor climate and air quality. *Environmental Research*, *243*, 117804. https://doi.org/10.1016/j.envres.2023.117804

[105] *Volatile Organic Compounds' Impact on Indoor Air Quality*. (2024, August 13). Environemtal Protection Agency. https://www.epa.gov/indoor-air-quality-iaq/volatile-organic-compounds-impact-indoor-air-quality

[106] Zhao, J., Uhde, E., Salthammer, T., Antretter, F., Shaw, D., Carslaw, N., & Schieweck, A. (2024). Long-term prediction of the effects of climate change on indoor climate and air quality. *Environmental Research*, *243*, 117804. https://doi.org/10.1016/j.envres.2023.117804

[107] *Health Effects of Ozone Pollution.* (2025, March 13). Environmental Protection Agency. https://www.epa.gov/ground-level-ozone-pollution/health-effects-ozone-pollution

[108] Kenny, G. P., Tetzlaff, E. J., Journeay, W. S., Henderson, S. B., & O'Connor, F. K. (2024). Indoor overheating: A review of vulnerabilities, causes, and strategies to prevent adverse human health outcomes during extreme heat events. *Temperature*, *11*(3), 203-246. https://doi.org/10.1080/23328940.2024.2361223

with chronic illnesses, and low-income residents who may lack access to air conditioning or energy-efficient housing. The consequences of extreme heat are not only physical but also mental. Studies have shown that high indoor temperatures are linked to increased sleep disruption[109] which can lead to higher rates of anxiety and irritability while also expatiating preexisting conditions[110].

Ithaca's older housing stock, much of which was not designed to withstand extreme heat, often lacks sufficient insulation, ventilation, or cooling infrastructure. Indoor temperatures in such buildings can exceed outdoor temperatures, creating potentially dangerous living conditions[111]. Compounding the issue, cooling centers may be inaccessible for those with limited mobility or transportation options[112], and energy costs associated with air conditioning are a significant barrier for many households[113]. This may be especially prominent in Ithaca since as of 2024, 28.4% of Ithaca's population is living in poverty[114]. Energy costs are also rising. In 2023, the New York State Public Service Commission approved a three-year rate plan for NYSEG, which includes a 62% increase in electricity rates and a 17.8% rise in gas delivery rates, which further strains the affordability of heating and cooling[115].

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[109] Obradovich, N., Migliorini, R., Mednick, S. C., & Fowler, J. H. (2017). Nighttime temperature and human sleep loss in a changing climate. *Science advances*, *3*(5), e1601555. https://doi.org/10.1126/sciadv.1601555

[110] Saghir, Z., Syeda, J. N., Muhammad, A. S., & Balla Abdalla, T. H. (2018). The Amygdala, Sleep Debt, Sleep Deprivation, and the Emotion of Anger: A Possible Connection?. *Cureus*, *10*(7), e2912. https://doi.org/10.7759/cureus.2912

[111] Teare, J., Mathee, A., Naicker, N., Swanepoel, C., Kapwata, T., Balakrishna, Y., du Preez, D. J., Millar, D. A., & Wright, C. Y. (2020). Dwelling Characteristics Influence Indoor Temperature and May Pose Health Threats in LMICs. *Annals of global health*, *86*(1), 91. https://doi.org/10.5334/aogh.2938

[112] Nayak, S. G., Shrestha, S., Sheridan, S. C., Hsu, W. H., Muscatiello, N. A., Pantea, C. I., Ross, Z., Kinney, P. L., Zdeb, M., Hwang, S. A., & Lin, S. (2019). Accessibility of cooling centers to heat-vulnerable populations in New York State. *Journal of transport & health*, *14*, 10.1016/j.jth.2019.05.002. https://doi.org/10.1016/j.jth.2019.05.002

[113] Ortiz, L., Gamarro, H., Gonzalez, J. E., & McPhearson, T. (2022). Energy burden and air conditioning adoption in New York City under a warming climate. *Sustainable Cities and Society*, *76*, 103465. https://doi.org/10.1016/j.scs.2021.103465

[114] *U.S. Census Bureau quickfacts: Ithaca City, New York.* (2024). United States Census Bureau. https://www.census.gov/quickfacts/fact/table/ithacacitynewyork/PST040224

[115] Dougherty, M. *Josh Riley Launches Investigation into NYSEG Over Rate Hikes and Transparency Complaints.* (2025, April 17th). Ithaca.com. https://www.ithaca.com/news/ithaca/josh-riley-launches-investigation-into-nyseg-over-rate-hikes-and-transparency-complaints/article_a85f6c87-0f0d-4927-8369-9d02d324c092.html

Flooding & Fungi

Ithaca's vulnerability to climate change is evident in its significant flood risk, which poses secondary challenges to indoor air quality and public health. Flood events, intensified by climate change, can lead to water intrusion in buildings, resulting in damp indoor environments that promote mold growth and the accumulation of indoor air pollutants[116]. These conditions can affect respiratory health and exacerbate chronic health conditions among residents[117]. Important factors encouraging mold growth after floods include floodwater depth, bathroom ventilation, roof age, structural sealing, and sunlight exposure through windows[118]. Flooding can also lead to disease outbreaks, drowning, and negative mental health impacts[119].

Data indicates that Ithaca is particularly at risk due to its low elevation and runoff from surrounding bodies of water[120]. As of 2025, 54.7% of properties are exposed to flooding, a figure expected to rise to 55.4%[121] over the next 30 years, compared with 21.7% in Tompkins County overall[122]. Flooding can damage building structures and HVAC systems, further diminish indoor air quality and increase exposure to contaminants that may lead to asthma, allergies, and other respiratory issues. Vulnerable populations, including low-

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[116] *Climate Change Impacts on Air Quality*. (2025, March 27). Environmental Protection Agency. https://www.epa.gov/climateimpacts/climate-change-impacts-air-quality

[117] *Climate Change Impacts on Air Quality*. (2025, March 27). Environmental Protection Agency. https://www.epa.gov/climateimpacts/climate-change-impacts-air-quality

[118] Pakdehi, M., Ahmadisharaf, E., Azimi, P., Yan, Z., Keshavarz, Z., Caballero, C., & Allen, J. G. (2025). Modeling the latent impacts of extreme floods on indoor mold spores in residential buildings: Application of machine learning algorithms. *Environment International*, *196*, 109319. https://doi.org/10.1016/j.envint.2025.109319

[119] Ohl, C. A., & Tapsell, S. (2000). Flooding and human health. *BMJ (Clinical research ed.)*, *321*(7270), 1167–1168. https://doi.org/10.1136/bmj.321.7270.1167

[120] Kreitinger, L. (2023, April 29). *Vital signs of change in our watershed: FEMA flood hazard areas expand as water rises.* Cayuga Lake Watershed Network. https://www.cayugalake.org/vital-signs-of-change-in-our-watershed-fema-flood-hazard-areas-expand-as-water-rise s#:~:text=The%20number%20of%20residences%20and,in%20a%20flood%20hazard%20areas

[121] *Ithaca, NY Flood Map and climate risk report.* (2025). First Street Technology. https://firststreet.org/city/ithaca-ny/3638077_fsid/flood.

[122] *Tompkins County, NY Flood Map and climate risk report. (*2025). First Street Technology. https://firststreet.org/county/tompkins-county-ny/36109_fsid/flood.

income residents, older adults, and non–English speaking communities, are disproportionately affected due to limited resources for flood mitigation and subsequent remediation of indoor environments[123].

Codes & Resiliency

As climate change intensifies, Ithaca faces growing public health and safety risks tied to building codes and infrastructure that may not be equipped to handle emerging environmental stressors. Rising temperatures, increased precipitation, and more frequent extreme weather events, including heat waves, flooding, and winter storms, pose threats to the structural integrity of buildings and the safety of their occupants[124]. Code updates that reflect current and projected climate realities, could play a significant role in decreasing vulnerability to hazards such as mold exposure, indoor overheating, poor air quality, and physical injury from building failures[125].

Older housing stock, which makes up a substantial portion of Ithaca's buildings (Mendizabal, 2024), is particularly at risk. Since most of these structures were built 30 or more years ago (Mendizabal, 2024), they were not built with energy efficiency or climate resilience in mind, making them ill-suited for prolonged heat or cold exposure and vulnerable to water damage from extreme precipitation ("Quality of", n.d.). These conditions disproportionately impact low-income residents, who often live in older buildings and may lack resources for retrofits or emergency repairs ("Quality of", n.d.). Health consequences can include heat-related illness, respiratory conditions ("Environment, Climate" n.d.), and displacement during climate-related disasters.

At the same time, the City of Ithaca's bold steps toward climate action in its Energy Code represent a protective factor for building stock climate resilience. The 2023 Ithaca Energy Code Supplement requires new buildings and major renovations to achieve substantial greenhouse gas reductions through a point-based system that prioritizes electrification, energy efficiency, and renewable energy (Ithaca Energy Code Supplement, 2023). The

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[123] Tompkins County Resiliency and Recovery Plan. (2022, July). Tompkins County https://www.tompkinscountyny.gov/files/assets/county/v/1/planning-amp-sustainability/documents/climate-adaptation/rrp_full_plan_updated_9_27_22.pdf

[124] Climate Change Impacts on the Built Environment. (2025, March 25). Environmental Protection Agency. https://www.epa.gov/climateimpacts/climate-change-impacts-built-environment

[125] Rajkovich, N. B., Brown, C., Azaroff, I., Backus, E., Clarke, S., Enriquez, J., Greenaway, B., Holtan, M. T., Lewis, J., Ornektekin, O., Schoeman, L., & Stevens, A. (2024). New York State Climate Impacts Assessment Chapter 04: Buildings. Ann NY Acad Sci., 1542, 214–252. https://doi.org/10.1111/nyas.15200

emissions-saving upgrades that are encouraged in the updated energy code can also reduce a building's climate risk: heat pumps provide effective cooling during extreme heat events (Tan & Fathollahzadeh, 2021), high-performance buildings maintain habitable conditions during power outages while preventing moisture damage (White & Wright, 2020), and renewable energy systems improve resilience by maintaining critical functions during grid disruptions from extreme weather (Xu et. al, 2024). While room for improvement still exists, particularly in relation to aging building stock and extreme weather events, the City of Ithaca's existing building code does promote climate resilience.

Preventative and Acute Care Access

Extreme weather events may disrupt transportation networks, damage healthcare facilities, and strain emergency services, limiting residents' access to both preventative care such as acute care services (Butsch, 2023), routine checkups, and treatment for heatstroke or injuries from severe storms. Vulnerable populations, such as the elderly (who make up 8.7% of the City of Ithaca's population) ("U.S. Census", 2024), individuals with chronic illnesses, and low-income communities, are particularly at risk of experiencing delayed or inadequate medical care during these events ("Climate Change and the Health", 2025). In addition to the physical damage to healthcare infrastructure, climate-induced stressors, such as heat and air pollution, can exacerbate pre-existing health conditions, increasing the demand for acute care services (Martins, 2024). Hospitals and clinics may become overwhelmed (Martins, 2024), particularly in smaller communities like Ithaca where healthcare resources are already limited. Events such as flooding, thunderstorms, and windstorms can also impact an individual's abilities to seek care as conditions outside might be too dangerous, or roads could be blocked.

Racial Equity

The risks associated with climate change, as discussed throughout these sections, are further compounded by race and class, making race and socioeconomic status a key predictor of both exposure to risk and strength of recovery from disaster[126]. The interconnections between climate and environmental risk and race have been studied for decades, arguably beginning with Dr. Robert Bullard, considered the Father of Environmental Justice, in the late 1970s and early 1980s. Regrettably, despite decades of research on risk exposure for BIPOC populations, the entrenchment of systemic racism is our existing systems continues to this day, and Black and Latinx populations remain at the

[126] Cardona, O, et al. *Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation: Special Report of The Intergovernmental Panel on Climate Change.* Cambridge University Press (2012).

highest risk for health and environmental vulnerabilities. This is why the City of Ithaca's climate action plan considers climate change an injustice accelerator, or risk multiplier, for racial minorities and recognizes that addressing racial inequities within the silo of climate change will produce limited results. Racial injustice needs to be addressed at its systemic root, first by understanding the complexities and compounding nature of exposure.

The City of Ithaca recognizes that the history of racial injustice and its systemic causes are both complex and wide-reaching, and therefore cannot be covered in their entirety in this document. The section below represents some of the underlying causes and results of racial inequity and its intersection with climate change, but it is not exhaustive.

Sustainability staff have also made the intentional decision to create a Racial Equity sector of the climate action plan to underscore its importance in climate change mitigation and resilience programming design and implementation. However, it is important to acknowledge that equity is deeply interconnected to issues related to housing, public health, labor, emergency response, and energy. There is widely available and accepted research documenting race as a predeterminant of various dangerous outcomes, which underscores the importance of a whole-system approach to climate and urban planning.

Redlining

In the 1930s, the U.S. government created homeownership programs that systematically denied financial services, like mortgages and insurance loans, to "undesirable" neighborhoods and populations. Terms like "undesirable", "subversive", and "inharmonious" were used to describe communities of color, particularly Black communities, by the Home Owners' Loan Corporation (HOLC), a program of President Roosevelt's administration[127]. This practice of denying services to communities of color resulted in the concentration of racial minority homeowners within certain neighborhoods and was eventually banned by the Fair Housing Act in 1968. "Shirley Mann-Smith says the problem was especially pressing in the Southside because 'the area was redlined by the banks. Tehy didn't want to lend money to people they thought couldn't repay it; all they could see was money going to pot. We had fifty or sixty abandoned houses here because even if the owners could repay a loan, the bank wouldn't finance them.'"[128] However, the legacy of redlining over the 30 years it was active remains today.

For example, research shows that neighborhoods and communities with a history of redlining are disproportionately exposed to extreme heat. While there is variability in the

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[127] Nelson, R. et al. *Mapping Inequality: Redlining in New Deal America*. University of Richmond (May 2025).
[128] Sisler, C. *Ithaca's Neighborhoods: the Rhine, the Hill, and the Goose Pasture* (1988). DeWitt Historical Society of Tompkins County. Pg. 108.

prevalence of extreme heat risk across formerly redlined neighborhoods, researchers found that 94% of studied areas display consistent city-scale patterns of elevated land surface temperatures in formerly redlined areas relative to their non-redlined neighbors by as much as 7 °C, with a U.S. average of 2.6 °C difference [129].

Heat

The effects of redlining reached farther than simply concentrating BIPOC populations in geographic areas; it also resulted in chronic underinvestment in public services in those neighborhoods and the frequent siting of environmental hazards nearby. For example, research from the Science Museum of Virginia on 108 U.S. cities found that previously redlined neighborhoods experienced extreme heat waves more acutely than other neighborhoods[130]. The reason for this is complex, as there are several factors that influence why some neighborhoods experience more intense heat. Previously redlined neighborhoods received fewer public funds for infrastructure that could decrease heat, like shade-providing trees, parks, and green space, while heat-trapping infrastructure, like concrete and asphalt, are rampant[131]. The resulting effect is redlined neighborhoods experiencing heat waves 5 degrees F hotter on average than other neighborhoods in the same city[41]. Compounding heat exposure, previously redlined neighborhoods typically have housing without air conditioning and lack cooling centers and/or bus service, meaning escaping dangerous heat waves is more difficult in redlined communities. In housing units where air conditioning is available, they are often expensive-to-run window units, which, in rental housing, landlords are under no obligation to repair or replace[132]. Keeping up with the ”energy burden”, or the percentage of household income spent on energy bills, in buildings subject to chronic underinvestment is both expensive and impacts other areas of healthy, daily life by decreasing funds available for other necessities like health care, healthy foods, transportation, and childcare. Recent utility rate hikes exacerbate this issue, with local electricity and natural gas supplier, New York

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[129] Hoffman, J., Shandas, V., Pendleton, N. *The Effects of Historical Housing Policies on Resident Exposure to Intra-Urban Heat: A Study of 108 US Urban Areas* (3 January 2020). Climate, 8(1), 12. https://doi.org/10.3390/cli8010012

[130] Hoffman, J., Shandas, V., Pendleton, N. *The Effects of Historical Housing Policies on Resident Exposure to Intra-Urban heat: A Study of 108 US Urban Areas*. Climate (January 13 2020).
[131] Cimons, M. *How redlining makes communities of color mroe at risk of deadly heatwaves.* Public Broadcasting Service (January 23 2020).
[132] Newsome, M. *Discrimination Has Trapped People of Color in in Unhealth Urban Heat Islands.* Nature (20 September 2023).

State Electric and Gas (NYSEG), increasing rates 62% since 2023 and a request to increase rates again by 31% by 2027. Further, uncertainty remains on whether federal financial assistance for residential energy bills will remain available into the future, as threats to the Low-Income Home Energy Assistance Program (LIHEAP) have persisted throughout 2025. A final decision on the future of LIHEAP is expected in September following Congressional appropriations. In Ithaca, this presents an unsettling future. The Census estimates 49.3% and 59.9% of Latinx and Black Ithacans' household earnings fall below the poverty line [133]. Energy burden across low-income groups in Tompkins County, the most granular data available, varies widely, with households earning 60% area median income (AMI) experiencing as low as 4% burden and the lowest-income earners experiencing 15% burden [134]. The American average energy burden is approximately 6%.

Disaster & Recovery

Extreme heat is the most probable and recurring hazard faced by communities; however, the risk of natural disaster still exists, and neighborhoods previously subject to redlining face particular dangers associated with this ugly history. The vulnerabilities to disaster are twofold: not only are previously redlined neighborhoods more likely to experience a disaster, they are also least able to recover from climate shocks quickly, creating an even wider wealth gap with surrounding communities.

A recent study released by Redfin analyzed current and historic maps across 38 major American cities, ultimately finding a positive correlation between HOLC neighborhood scores and the percentage of homes that fell within a 100-year flood zone. In other words, researchers found that, nationally, there are $107 billion worth of homes at high risk for flooding located in what HOLC previously deemed "undesirable" neighborhoods, while in formerly "desirable" neighborhoods, only $85 billion worth of homes are at high risk[135].

While Ithaca HOLC maps are unavailable and thus this granularity of analysis is not possible at the local level, the underlying methodology FEMA utilizes to assess risk shows a consistent pattern in the City of Ithaca. In 2023, FEMA updated its risk assessment methodology and source data to include evaluation of social vulnerability and resilience in the event of a flood event[136]. These social metrics consider four indicator categories to

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[133] U.S. Census Bureau. *Poverty Status in the Last 12 Months* (2023). American Communities Survey 5-Year Estimates Subject Tables, Table 1701. Data.census.gov
[134] Low-Income Energy Affordability Tool. *Energy Burden for Tompkins County (2023).* U.S. Department of Energy. https://www.energy.gov/scep/slsc/lead-tool
[135] Katz, L. *A Racist Past, a Flooded Future: Formerly Redlined Areas Have $107 Billion Worth of Homes Facing High Flood Risk – 25% More Than Non-Redlined Areas.* Redfin (14 March 2021).
[136] *Updates | National Risk Index*. 2023. Federal Emergency Management Agency. https://hazards.fema.gov/nri/updates.

determine the overall vulnerability score of a neighborhood: socio-economic status, household characteristics, racial and ethnic minority status, and housing type and transportation[137]. When solely reviewing the Social Vulnerability Index (SVI) of the City of Ithaca, the most socially vulnerable communities are the Southside, Northside, and West End neighborhoods – all of which are also within the Special Flood Hazard Area[138]. Further, an internal analysis of 2022 median home values by city staff revealed that the same neighborhoods and census tracts had the lowest assessed value per parcel[139] and the highest density of Black and Hispanic and/or Latinx individuals living in those neighborhoods[140] presently and historically. This trend is reflected nationally, with an $80,000 median difference in home value between Black and white Americans[141]. Further, research has shown that Black homeowners remain concentrated in formerly redlined neighborhoods[142].

On top of the acute disaster risk and the relative social and economic vulnerability of BIPOC populations that make communities more susceptible to disasters, significant racial disparity also exists in disaster recovery efforts. Due to several compounding factors, including the complex and ugly history of redlining, chronic undervaluation of Black-owned properties and neighborhoods, and underdevelopment, Black communities and neighborhoods receive significantly less relief assistance, take longer to recover, and are less likely to relocate to safer areas after a disaster. The underlying reason for all three is the significant discrepancy in relief assistance, ultimately stemming from the municipal scars of redlining and the economic turmoil that follows on the heels of a disaster.

For decades, the policies implemented by FEMA have inadvertently perpetuated the inequities established by the 1930s HOLC maps. Research by major research institutions, policy advocacy groups, and even the federal government has confirmed that agency practices contributed to systemic racism. In late 2020, prior to President Biden taking office, the FEMA National Advisory Council (NAC) shared a report with then-Administrator

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[137] *Social Vulnerability Index.* 22 July 2024. Center for Disease Control: Geospatial Research, Analysis, and Services Program (GRASP). https://www.atsdr.cdc.gov/place-health/php/svi/index.html
[138] *SVI Interactive Map.* 2022. Center for Disease Control: Geospatial Research, Analysis, and Services Program (GRASP). https://www.atsdr.cdc.gov/place-health/php/svi/svi-interactive-map.html
[139] Aslanis, R. and Bijl, A. *Climate Justice Communities Data Compilation 2022 Findings*. 31 October 2022. City of Ithaca GIS Program.
[140]
[141] Bhutta, N., et al. *Disparities in Wealth by Race and Ethnicity in the 2019 Survey of Consumer Finances.* 28 September 2020. United States Federal Reserve: FEDS Notes. https://www.federalreserve.gov/econres/notes/feds-notes/disparities-in-wealth-by-race-and-ethnicity-in-the-2019-survey-of-consumer-finances-20200928.html
[142] Zonta, M. *Racial Disparities in Home Appreciation: Implications of the Racially Segmented Housing Market for African Americans' Equity Building and the Enforcement of Fair Housing Policies.* 2019. Center for American Progress. https://www.americanprogress.org/article/racial-disparities-home-appreciation/

Gaynor detailing FEMA's inequitable practices, "{Recovery programs} They provide an additional boost to wealthy homeowners and others with less need, while lower-income individuals and others sink further into poverty after disasters"[143]. Two specific practices are largely to blame: the 50% FEMA Rule Appraisal and FEMA's distribution of Hazard Mitigation funds; both are described briefly below.

Prevention

As Ithaca is acutely aware, the ability of communities to prevent disasters can have long-term socio-economic benefits that extend beyond preserving infrastructure, like preventing further gentrification, enabling aging-in-place, and preserving the cultural fabric of pocket communities. Often, a community's ability to implement disaster prevention measures hinges on securing multi-million-dollar grants facilitated by FEMA, like the flood mitigation project in the City of Ithaca. The challenge with securing these large grants is threefold and continuously puts lower-income and communities of color at a disadvantage. At the forefront, the application process for large grants can be a deterrent for many communities, as federal applications are often tedious, time-consuming, and require specialized expertise and planning. In municipalities already struggling financially, these hurdles can be too costly to bear and prevent either successful funding or application altogether. Further compounding the issue, hazard mitigation grants require a local match to disburse funds, which many communities with limited capital funds, like Ithaca, simply cannot afford.

Finally, and perhaps most importantly, the distribution of mitigation funds is particularly concerning and pervasive. Because demand for hazard mitigation funds far exceeds their availability, FEMA has implemented a methodology to determine how to distribute funds based on a cost-benefit analysis. Under FEMA's cost-benefit methodology, benefits are equal to the anticipated damages without mitigation funds, minus estimated damages with mitigation measures[144] . The analysis considers public infrastructure and services, physical damage, temporary housing costs, loss of business income, and similar costs that would otherwise be avoided through disaster prevention measures. However, in communities without high business production, low home valuation, and aging

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[143] *National Advisory Council Report to the FEMA Administrator*. November 2020. U.S. Department of Homeland Security. https://cdn.prod.website-files.com/635adac789d0118ea58622fb/635adac789d01139788625db_National%20Advisory%20Council%20Report%20to%20the%20FEMA%20Administrator%202020.pdf

[144] *Engineering Principles and Practices for Retrofitting Flood-Prone Residential Structures. Appendix B: Understanding the FEMA Benefit-Cost Analysis Process.* 2012. U.S. Department of Homeland Security. https://data.wvgis.wvu.edu/pub/RA/_resources/Archive/Mitigation/NonStructural/fema259_complete_rev.pdf

infrastructure - all lingering symptoms of redlining and limited capital funds - the monetary value and representation of “benefit“ of protecting those resources is categorically undervalued. Thus, by FEMA’s own methodological design, low-income and Black communities receive fewer hazard mitigation funds, making them more vulnerable to disasters and less likely to recover. While we cannot determine if this methodology ultimately impacted the amount of funding Ithaca has or will receive through FEMA grants, it is worth noting that the structural issues created by redlining nearly 100 years ago still persist today and may ultimately affect the capacity for resilience in the future.

Recovery

In the unfortunate scenario that a community experiences a natural disaster, equity concerns also proliferate within FEMA’s allocation of recovery funds. The most common aid provided by FEMA, the Individuals and Households Program (IHP), assists uninsured or under-insured households affected by disaster with financial assistance and direct services. Assistance includes funds for temporary housing or to repair owner-occupied homes. However, like most FEMA programs, the demand for assistance far exceeds the fund balance, which requires FEMA to use an allocation methodology to distribute resources.

Two primary themes are revealed when assessing the distribution of FEMA funds: Black households face disproportionate obstacles to receiving resources, and renters receive considerably fewer funds than property owners. There is also significant overlap between the themes, with 56% of Black Americans renting their homes.

One of the allocation methodologies fueling the disproportionate distribution of funds is the 50% Rule Appraisal, which assigns disaster aid based on property values prior to a weather event. Research has found that homes in Black neighborhoods are valued approximately 21-23% below what they would be in non-Black neighborhoods[145]. The reason for undervaluation is systemic, but can be traced back to the history of redlining and how many appraisers were trained to execute their job functions, according to 2022 testimony from Lisa Rice, President of the National Fair Housing Alliance[146]. Further, in November 2022, the New York Times reported that racial inequality in home appraisals has increased 75% in the past decade[147]. With the intensity and frequency of storms increasing

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[145] Rothwell, J. and Perry, A. *How racial bias in appraisals affects the devaluation of homes in majority-Black neighborhoods.* 5 December 2022. Brookings Institution. https://www.brookings.edu/articles/how-racial-bias-in-appraisals-affects-the-devaluation-of-homes-in-majority-black-neighborhoods/

[146] Rice, L. *Devalued, Denied, and Disrespected: How Home Appraisal Bias and Discrimination Are Hurting Homeowners and Communities of Color.* 29 March 2022. National Fair Housing Alliance: Testimony before the U.S. House Committee on Financial Services.

[147] Kamin, D. *Widespread Racial Bias Found in Home Appraisals.* 2 November 2022. New York Times.

due to climate change, communities can expect damage to public and personal property. The ability to recover from storms and other natural disasters is a necessary component of fostering resilient communities, and current systemic structures are prohibitively complex or burdensome, preventing communities of color from equally accessing available resources.

Wealth & Community Building

Economic security has long been defined by the ability to obtain a good-paying job, and has been the foundation of our country's political and social framework. Often, steady career growth begins early, and has positive implications for an individual's future health, economic status, and academic outcomes. However, even in times of nation-wide economic success, communities of color still face more challenges to landing stable jobs than their white counterparts[148]. Further, labor market research reveals that even despite recent successes in increasing workers' access to jobs, and actually securing more jobs, specific "labor market outcomes – including higher unemployment and fewer good jobs – continue to be worse for African American workers and their families"[149]. "Good jobs" can be somewhat vague, however Black workers have historically earned lower wages than white workers and have fewer employer-provided benefits, increasing overall costs of living. A study by the Employee Benefits Research Institute in 2018 found that Black American workers were 14% less likely than white workers to have any type of retirement plan through their employer, making it difficult for communities of color to build community and generational wealth[150]. For these reasons, it is essential to incorporate workforce programming rooted in equity and wealth building, and that removes the systemic hurdles that have historically prevented communities of color from building wealth.

While family and individual wealth-building within communities of color are important, there are other indicators of equity and access that deserve equal attention in program planning. Community-building, a term that is frequently used to describe the non-physical

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148 Reid, L. *Integrating Economic Dualism and Labor Market Segmentation: The Effects of Race, Gender, and Structural Location on Earnings, 1974-2000* (2016). The Sociological Quarterly, 44(3). https://doi.org/10.1111/j.1533-8525.2003.tb00539.x

149 Weller, C. *African American Face Systematic Obstacles to Getting Good Jobs* (2019). Center for American Progress. https://www.americanprogress.org/article/african-americans-face-systematic-obstacles-getting-good-jobs/

150 Copeland, C. *Current Population Survey: Checking in on the Retirement Plan Participation and Retiree Income Estimates* (2019). Employee Benefits Research Institute. https://www.ebri.org/content/current-population-survey-checking-in-on-the-retirement-plan-participation-and-retiree-income-estimates

aspects of development – including organizational capacity building, the changing of decision-making systems, or resource allocation processes – is one such example. Historically, exclusionary practices, like over policing in communities of color, underinvestment in public infrastructure, or targeting of poor communities for undesirable development has eroded trust between the public and government. Research conducted by Building Bridges, a project of the Dorothy Cotton Institute, confirmed in 2020 and 2021 that local communities of color in Ithaca also experienced various levels of distrust with local government. In order to repair the eroded trust created by generations of underinvestment and exclusion at varying levels of government, it is essential that local governments begin to rebuild systems previously used as weapons of oppression. Further, communities of color must not only have input, but lead these efforts to ensure that true accessibility, inclusion and equity are represented both in process and implementation.

Emergency Response

Emergency preparedness is perhaps the most visible component of climate planning. However, we advocate that emergency response does not begin or end with onsite crisis management. A thoughtful emergency plan includes robust risk assessments, resource inventories, delegation, onsite management, and recovery efforts. The sections below do not attempt to fill any of these responsibilities; rather, they point out some of the risks present in the City. To appropriately develop an emergency response plan, the City should work in consultation with an Emergency Manager and existing first responders to complete comprehensive risk assessments based on present and future conditions and capacity.

Sustained Blackouts

With climate change expected to increase the frequency and severity of storms and extreme temperature swings, the risk of sustained blackouts can also be expected to increase over time. The City's ambitious electrification goals, while a powerful step towards climate change mitigation, can also potentially increase the risk of sustained blackouts by increasing reliance on electricity for power generation (*Electrify Ithaca*, n.d.; Majowicz et al., 2024). Communication and mobility can be life-or-death in emergency situations, including sustained blackouts. Because of this, ensuring the consistent performance of telecommunications systems and the smooth movement of traffic during blackouts in the City of Ithaca is a critical task.

Telecommunications Systems

Telecommunications systems have a massive impact on everyday life and are heavily dependent on electricity. Sustained blackouts can cause cascading failures in telecommunication systems, from large-scale infrastructure like towers to individual cell phones and radios, if backup power is unavailable (Kraussman et al., 2017; Pescaroli et al., 2017). In recent years, emergency responders have become increasingly reliant on electricity-dependent telecommunications systems, particularly cell phones and two-way radios (Kraussman et al., 2017). Beyond emergency responders, telecommunications systems are also crucial to disseminate health and safety information to the public, such as boil water advisories or shelter-in-place orders. Thus, sustained blackouts present a major risk to emergency response efficacy, such as delays in ambulance dispatch, patient triage, and transport to healthcare facilities, resulting in increased morbidity and mortality, particularly for time-sensitive conditions like cardiac and respiratory events or trauma.

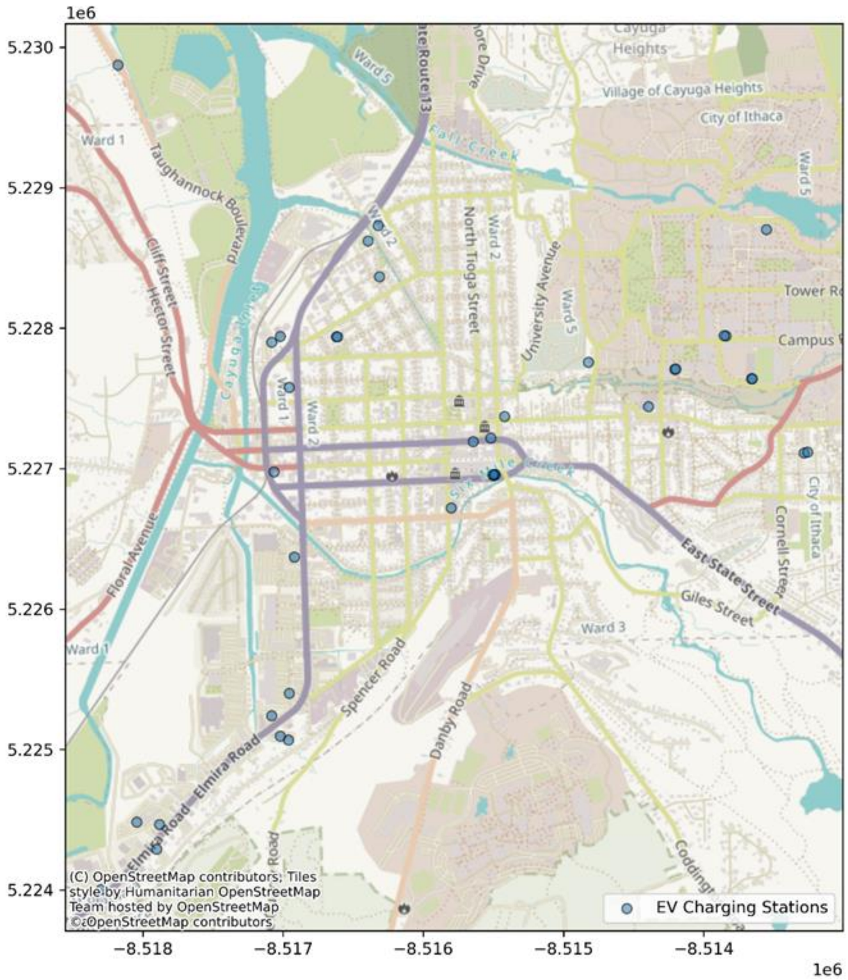
The City of Ithaca contains 176 registered cell towers; the largest number of these towers are registered by Cornell University, Tompkins County Department of Emergency Response, and Saga Communications, LLC, respectively (Cell Tower Maps, n.d.). The high number of cell towers in the City of Ithaca is a protective factor in sustained blackout scenarios. In the event of a sustained blackout, having network redundancy through multiple towers means that if one tower fails, others can potentially maintain coverage in the area. Additionally, cell towers do have backup power that can last for up to 48 hours. During a sustained blackout that exceeds 48 hours of power loss, however, the high level of dependence of cell towers on electricity presents a major risk to critical communications.

Radio communications are another important component of telecommunication systems. Within the City of Ithaca, there are four FCC-registered radio towers and 32 radio stations within close listening range, including several key community radio stations such as WRFI, WICB, and WVBR (Radio-Locator, n.d.). As with cell towers, redundancy in radio infrastructure and stations represent protective factors during sustained blackouts. In previous disaster scenarios in several different contexts, amateur radio has been a major lifeline for emergency communication after other systems failed (Gill, 2019). At the same time, radio telecommunications systems are also heavily dependent on electricity and face additional issues with technical failures (such as software crashes and equipment malfunctions), physical damage, network congestion, and limited coverage during disaster scenarios (Debnath et al., 2022). Ensuring backup power and redundancy within radio systems is key for emergency planning.

Traffic

Sustained blackouts can also impact mobility and traffic patterns. According to 2023 data from the Ithaca-Tompkins County Transportation Council, Route 13 at 3rd Street had the highest overall traffic volume, with additional heavy traffic points along 13 at Elmira Road, Fulton Street, and Meadow Street. Additional roadways with high traffic volume included Elmira Road, Spencer Street, and Route 79 (2023 Traffic Count Report, 2024). Sustained blackouts can cause failures in traffic lights and light control infrastructure, exacerbating traffic congestion and slowing response time for ambulances, fire trucks, law enforcement, and other emergency responders (Lu et al., 2020). Disruptions to traffic light systems can also lead to accidents and worsened air quality from car idling. Concurrent emergencies such as wildfires and storms could further worsen traffic congestion during sustained blackouts.

Households can also face risk from traffic disruption during emergency situations, as congestion also slows down evacuation from disaster-impacted areas. Owners of electric vehicles (EVs) must also consider the logistics of charging essential fleet vehicles during evacuations. The City of Ithaca has a total of 38 EV charging stations (Alternative Fuels Data Center, 2025). Of these stations, 4 are free of charge and 4 stations are DC fast chargers, with a combined total of 20 fast charging ports (Alternative Fuels Data Center, 2025). Especially in an evacuation scenario where a high volume of drivers can be expected to use similar routes to exit the city, the capacity of the EV charging network may not be enough to meet demand. At the same time, in rare cases, damage to EV charging infrastructure could potentially lead to fires (Electric vehicles and flooding, 2024). For these reasons, the location of electric vehicle charging infrastructure is a key consideration in emergency response planning.



Flooding

As climate change increasingly impacts the City of Ithaca, flood severity and frequency can be expected to increase. More flooding presents a significant risk to the health, safety, and property of Ithaca residents. Flood conditions can rapidly transform streets into waterways, damage homes and businesses, and create hazardous conditions that require complex emergency response.

Evacuation

Timely evacuation during a major flood can mean the difference between life and death. Refusing or being unable to evacuate isolates people physically, presents major challenges for caring for vulnerable loved ones, and may pose grave physical danger to those who shelter in place (Haynes et al., 2018). Successful large-scale flood emergency response, including evacuation, relies on coordinated decision-making by local emergency management officials. The Department of Homeland Security recommends that municipalities plan for floods by creating evacuation zones based on hazard exposure, evacuate communities together to promote social support, disperse evacuees throughout the community to reduce the burden on individual locations, and communicate safe evacuation routes to community members (Evacuation and Shelter-in-Place Guidance, 2019). Planning for evacuations should also prioritize maintaining access to hospitals and other healthcare facilities. As it currently stands, neither the City of Ithaca nor Tompkins County has any identified flood evacuation zones or routes, despite a robust understanding of hydrologic flood risk in the area (Tompkins County Resiliency and Recovery Plan, 2022). Furthermore, Tompkins County has nearly 30 miles of major roadways exposed to the 1% annual chance flood event (Tompkins County Hazard Mitigation Plan-Flood, 2021). The lack of identified flood evacuation zones and routes may slow or prevent a coordinated response during a major flood event, which significantly increases community flood risk.

One group that faces unique challenges during flood emergency evacuations are community members with disabilities. In Ithaca, approximately 7.7% of the population under 65 has a disability, with another 8.7% aged 65 or older (*U.S. Census Bureau QuickFacts*, 2023). Municipal emergency departments across the country improve response by maintaining voluntary emergency contact lists for disabled individuals (U.S. Department of Homeland Security, 2025). Such registry programs help responders locate and assist people with disabilities during natural disasters. As of 2025, Tompkins County lacks this specific service for Ithaca residents despite providing general flood emergency assistance (Tompkins County Department of Social Services, 2025). Different disabilities also require tailored accommodations in emergency procedures.

Mobility-impaired people face many barriers in flood emergency evacuation situations, including loss of assistive devices and medical equipment, lack of access to transit, inaccessible emergency shelters, and difficulty evacuating high-rise buildings. Such barriers are worsened by floodwaters or debris blocking roads and shelters (Raja & Narasimhan, 2022). People with other disabilities also encounter significant barriers to accessing and receiving timely emergency response services during flood events. People with sensory disabilities (including blindness, deafness, and speech impairments) face challenges in accessing warnings, navigating unfamiliar environments, and communicating with emergency responders. Individuals with cognitive disabilities may struggle with understanding or remembering instructions and emergency contact information, making it difficult to follow evacuation instructions or access services during disasters (Raja & Narasimhan, 2022).

Within the City of Ithaca, major roadways identified by the county as common evacuation routes (including Routes 13, 79, 96, 34 and 89) can be expected to be flooded in the event of a major flood (Tompkins County Hazard Mitigation Plan-City of Ithaca, 2021; National

Flood Hazard Layer Map Viewer, 2025). Accessible information about safe alternative evacuation routes, emergency procedures, and shelter locations can save the lives of people with disabilities by expediting emergency response (Fox et al., 2007). This information can and should come in multiple formats, including but not limited to signage, route maps, audio announcements, braille materials, simplified text with visual aids, digital applications with screen reader compatibility, and direct outreach through community-based organizations serving people with disabilities in Ithaca (National Council on Disability, 2005). A lack of specified evacuation routes and outreach puts disabled community members further at risk to adverse impacts from floods and other natural hazards.

Another disproportionately vulnerable group during flood events is individuals experiencing homelessness. Flood evacuation for the homeless community presents challenges since they frequently live in marginal, flood-prone areas, often lack access to emergency warning systems, and have higher rates of chronic diseases that can be worsened by floods and wet weather (Every, 2016; Ramin & Svoboda, 2009). Additional health and socioeconomic burdens further complicate emergency response during floods for individuals experiencing homelessness. Poverty, substance abuse, and mental illness that are common in the homeless population can prevent people from engaging with emergency response efforts or services that are often complex and difficult to navigate, increasing risk (Every et al., 2019). Lastly, a lack of knowledge about the location and number of individuals in the homeless populations in Ithaca presents a major barrier to effective flood planning or evacuation.

The most recent Ithaca/Tompkins County Point in Time Count of the homeless population (from January 25th, 2024) counted 210 total homeless persons in the county, with 16 people totally unsheltered (U.S. Department of Housing and Urban Development, 2024). At the same time, other reporting places the population of the encampment located on the City-owned parcel in the southwest, at 20 to 60 people alone; these numbers may have changed since the city began clearing the area in May 2024 (Jordan, 2023; Lucas, 2024). The absence of comprehensive data on the locations and numbers of people experiencing homelessness creates a critical gap in flood evacuation planning, significantly hampering emergency responders' ability to locate and assist this vulnerable population during flood events.

Critical Services

Water supplies are vulnerable to damage from flood events. Severe flooding can worsen groundwater quality and destroy supply infrastructure such as pumping stations (Barbetta et al., 2022). Factors that contribute to the flood risk of municipal infrastructure include flood susceptibility, weather, population density, average road density, and average degree of imperviousness (Santos et al., 2020). One major protective factor in the City of Ithaca is the risk assessment that Tompkins County completed for its 2021 Hazard Mitigation Plan. This plan identified one wastewater pumping station within the City of Ithaca's 1% annual chance floodplain and an additional potable water tank and wastewater treatment facility in the City's 0.5% annual chance floodplain (Tompkins County Hazard Mitigation Plan-Flood, 2021, pp. 5.4.4-45-46). In a changing climate, the City's floodplains may expand in the future, posing more of a risk.

Flood events can make it more difficult to access hospitals and reduce the mobility of emergency responders like ambulances, the fire department, and law enforcement. The 2021 Hazard Mitigation Plan determined that the City of Ithaca can expect 9,604 tons of debris to be generated by the 1% annual chance flood event (Tompkins County Hazard Mitigation Plan-Flood, 2021, p. 5.4.4-42). Debris of this magnitude could realistically block roads and prevent members of the community from accessing hospitals or other critical facilities. According to the Hazard Mitigation Plan, the City of Ithaca had 3 critical facilities located in the 1% annual chance floodplain (1.5% of all critical facilities) and 30 critical facilities located in the 0.5% annual chance floodplain (14.9% of all critical facilities) as of 2020 (Tompkins County Hazard Mitigation Plan-Flood, 2021, pp. 5.4.4-45-46). Furthermore, modeling produced by City of Ithaca contractors shows that vehicles from one of the major emergency service providers in the City of Ithaca, Bangs Ambulance, may be unable to exit their facility's driveway in the event of a flood, and that all surrounding streets would be blocked by high water. Damage to critical facilities can contribute to cascading failures, worsen disaster impacts, and threaten community life safety (see "Preventative and Acute Care Access" in the Public Health section of this document for more information).

Extreme Temperatures

Climate change can influence temperatures at both extremes—heat and cold. Atmospheric warming increases the frequency and intensity of heat waves. At the same time, disruption of atmospheric patterns like the polar vortex can lead to more severe cold snaps in certain areas, creating unpredictable winter conditions despite an overall warming trend. As a lakefront city in the northeastern United States, the City of Ithaca must consider both extreme heat and extreme cold in its climate change adaptation policy.

Pushing Urgency

Heat waves kill more people annually than floods, tornadoes, and earthquakes combined (Adams-Fuller, 2023). Mere exposure to extreme heat can lead to heat stroke, which has cascading mental and physical impacts (Seltenrich, 2015). According to the County's 2021

Hazard Mitigation Plan, the Ithaca area may experience more frequent days over 90°F, potentially increasing from 10 to 23 days annually depending on the climate change scenario (Tompkins County Hazard Mitigation Plan-Extreme Temperatures, 2021, pp. 5.4.3-6-7). Extreme cold temperatures are also a dangerous natural hazard. Hypothermia is a major risk during cold snaps, and existing respiratory health problems are exacerbated by cold (Seltenrich, 2015). Extreme temperatures can also lead to other hazards, such as fires and drought with extreme heat and power outages with extreme cold, creating simultaneous emergencies.

Homeless Population

As mentioned previously, a lack of access to emergency warning systems, higher chronic disease rates, and inadequate data about the location of individuals experiencing homelessness make extreme temperatures a major risk to the life safety of people experiencing homelessness. Unsheltered individuals are at significant risk for developing hypothermia during cold snaps and hyperthermia during heat waves (Bezgrebelna et. al, 2021; Noor et. al, 2025). Emergency response for the homeless community remains a challenge during extreme temperature events. Physical, social, and psychological challenges can prevent the homeless community from accessing shelters during cold snaps or cooling centers during heat waves (Bezgrebelna et. al, 2021). While no specific data has been collected about the City of Ithaca, research indicates that during heat waves in particular, emergency department admissions of people experiencing homelessness may significantly increase (Noor et. al, 2025). At the same time, the same persistent challenges that prevent individuals experiencing homelessness from accessing shelters and cooling centers can also prevent them from seeking out necessary medical care (Koehler, 2021). Unsheltered individuals in the City of Ithaca represent one of the highest risk populations during extreme temperatures.

Health-Compromised Population

Extreme temperature can exacerbate existing health problems. People who are above the age of 65, have chronic diseases or drug and alcohol use disorders, experience social isolation, or live in areas with little tree cover are significantly more likely to be hospitalized or experience adverse health outcomes during extreme temperature events (Hess, 2023; Layton et. al, 2020; Seltenrich, 2015). Physical health factors, social connection, and the built environment all contribute to heat risk for health-compromised people in the City of Ithaca. Extreme cold also presents risk to health-compromised individuals. Many Tompkins County residents suffer from chronic diseases. Data from 2020-2022 indicates that Tompkins County had an age-adjusted chronic lower respiratory disease mortality rate

(33.2%) higher than the statewide rate (23.7%). On the other hand, the County had lower mortality and hospitalization rates than the state for asthma and all tracked cardiovascular diseases (*New York State Health Indicators by Race and Ethnicity, 2020-2022*, n.d.; *Tompkins County Health Indicators by Race and Ethnicity, 2020-2022*, n.d.). Targeting chronically ill populations in emergency response is crucial to saving lives during extreme temperatures.

Power Reliability

The reliability of electricity has implications that extend beyond daily household convenience. U.S. infrastructure like telecommunications, hospitals and medical facilities, the economy, and transportation all rely heavily on the stability of the grid and distribution infrastructure. Simultaneously, the U.S., New York State, and the City of Ithaca are rapidly moving toward increasing reliance on electricity and decreasing dependency on fossil fuels for both equity and environmental reasons. For outdated infrastructure unprepared to meet this increasing demand, this presents new strains on our electricity infrastructure and limitations to social and economic growth. These strains are only exacerbated by changes in weather patterns anticipated as a result of climate change, including shifts in seasonal peaks, natural disasters, and an increase in the number and severity of storms that can damage infrastructure.

In the event of a power outage, most localities are unprepared to sustain critical infrastructure beyond a couple of a days and, instead, rely on utility companies to restore power quickly. As recently as 2024, a severe thunderstorm disrupted power for thousands of Ithaca and Tompkins County residents for 72 hours. Because of the increasing reliance on electricity for our daily lives and for our most critical services, it is essential that the City think proactively about decentralizing and diversifying power resources to fill in the gaps in identified vulnerability gaps.

Centralization

The majority of electricity in the United States is generated at large-scale, centralized facilities that are connected through a network of high-voltage transmission lines and distributed to end-users. While there are many kinds of centralized facilities, most of them produce electricity through the combustion of fossil fuels, like natural gas. Ithaca's electricity network is somewhat unique, with approximately half of our electricity production coming from carbon-free sources like hydropower and nuclear[151]. While the centralized nature of power distribution is more efficient and economically beneficial for

[151] NYISO. *Power System Fundamentals.* 2020. New York Independent Systems Operator.

utility companies, it also presents unique challenges that increase vulnerability for customers. One such example is the power lost as electrical energy travels from the point it was generated to its end-use – also known as transmission loss. In New York, we assume a loss rate of just over 7%[152]. This loss rate represents excess greenhouse gas emissions, real power, and economic losses, and reflects one of the drawbacks of a principally centralized power system. However, there are other examples of downsides to centralized power.

Customer Cost & Choice

The costs associated with a centralized power system have both positive and negative implications. On the one hand, the costs of transmission and distribution infrastructure upgrades are extremely expensive. For example, NYSEG, Ithaca's electric grid operator, is forecasted to spend approximately $294.7M in electric and generation capital investment for calendar year 2025[153]. The U.S. Energy Information Administration estimates that across the U.S., spending by major utilities has increased 12% between 2003 and 2023 based on FERC data, likely due to the replacement of aging infrastructure, introduction of automated systems like smart meters and sensors, and new line installation[154]. On the surface, the cost of these necessary upgrades appears antithetical to customer cost control, but in reality, these costs are socialized across large service territories. NYSEG, for example, serves more than 40% of New York State and nearly 1,000,000 customers, meaning the $294.7M in capital costs is distributed across all 1M customers, theoretically saving residents money.

However, this socialization structure is not always fair. Major users, like data centers, require far more capital investment in the grid to accommodate their supersized electric loads. Hyperion, for example, Meta's latest datacenter project in Louisiana, is projected to have a peak demand of 5GW, about half the demand of New York City[155]. In this scenario, Meta is the sole driver of the costly upgrades required to operate the new facility. However, there is little transparency around Meta's financial responsibility for the upgrades versus

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[152] Hammer, H. *Projected Emissions Factors for New York State Grid Electricity.* August 2022. New York State Energy Research & Development Authority.
[153] NYSEG and RG&E. *NYSEG and RG&E Capital Investment Plan Quarterly Variance Report Q3 2025.* 17 November 2025. Submitted to the NY Public Service Commission as part of cases 22-E-0317, 22-G-0318, 22-E-0319, and 22-G-0320.
[154] U.S. Energy Information Administration and Federal Energy Regulatory Commission. *Grid infrastructure investments drive increase in utility spending over past two decades.* 18 November 2024. U.S. Energy Information Administration.
[155] Thomas, M. *These Data Center Are Getting Really, Really Big: Gigawatt-sized data centers are becoming the new normal.* 16 October 2025. Distilled Magazine.

other ratepayers[156]. It's scenarios like these, the introduction of major power consumers, that present challenges to areas with already stressed electrical infrastructure, like Ithaca and Tompkins County, raising both technical and ethical questions over who should pay and whose service will be prioritized.

Beyond the cost of necessary upgrades, the natural monopoly created by the utility industry and public policy also confines consumers to limited choice and often higher cost. Over the past several years, Tompkins County and other central New York residents have expressed growing frustration with ongoing billing errors, declining customer service, and increasing delivery rates. Multiple media outlets have covered both the frustration and the subsequent third-party audit ordered by the Public Service Commission[157],[158],[159]. While sole company ownership over transmission and distribution infrastructure can keep costs low by eliminating redundancies and increasing efficiencies and oversight, they offer few, if any, other options for ratepayers when frustration grows high enough to seek alternatives. Further, with fewer options available, customers are less likely to participate in the democratic process and structures that are built to protect ratepayers from exorbitant rate hikes, unnecessary tariffs, or bills issues.

Flexibility

A major drawback of centralized power is its low flexibility and lack of resilience. As both demand and energy input to the grid evolve and weather patterns change, centralized infrastructure is not designed to manage load variability and result in unreliable electrical service[160]. As cities like Ithaca continue to deploy renewable energy systems with grid interconnections, we are forced to grapple with the challenges associated with the innate intermittency of source generation, like the availability of solar or wind energy, and the common variability of temperature spikes, particularly in peak seasons. Transmission congestion of this nature impacts both the transmission system as a whole and individual end-users. For example, in high temperatures or in periods of high-demand, transmission lines can swell and sag beyond their usual range, damaging the infrastructure permanently[161]. In turn, additional investments in grid infrastructure are needed to both

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[156] Hawkins, D. *Construction on Meta's largest data center brings 600% crash spike, chaos to rural Louisiana*. 2022 November 2025. Louisiana Illuminator.
[157] Dougherty, M. *Josh Riley Launches Investigation into NYSEG Over Rate Hikes and Transparency Complaints*. 17 April 2025. Ithaca Times.
[158] Rovenolt, M. *NYSEG: Addressing shortfalls and customer complaints.* 15 March 2023. Tompkins Weekly.
[159] Vogel, M. *Audit of NYSEG Finds Billing Issues, Leadership Conflicts.* 12 June 2025. Ithaca Times.
[160] Kabir, E., et al. *Quantifying the impact of multi-scale climate variability on electricity prices in a renewable-dominated power grid (*2024). Renewable Energy. https://doi.org/10.1016/j.renene.2024.120013
[161] Joint Economic Committee. *How Renewable Energy Can Make the Power Grid More Reliable and Address RIsks to Electricity Infrastructure* (2024). U.S. Senate.

repair and reinforce transmission wires, raising costs for ratepayers. Further, when peak load variability extremes occur, like a very sunny, hot day, utility companies can practice service curtailment, which is the intentional reduction of power in an attempt to balance the grid. On the opposite end of the spectrum, when energy supply is overproduced and causes congestion, this can cause power surges, or sudden spikes in electrical voltage. Surges can permanently damage anything that is grid-tied, including home appliances, further raising costs for customers. There are load management solutions that can help with this variability and allow for the rapid deployment of renewables[162],[163]. Many of these solutions are included in the *Recommendations* section of this Climate Action Plan.

Resilience

Electrical Grid Outages

The following assessment is borrowed from the Tompkins County report *Navigating Electrical Outages: Proactive Steps for Today and Tomorrow's Electrified World*. The City of Ithaca thanks the County's Planning & Sustainability Office for their hard work in producing the report and for their permission to reproduce it here. The section from the County's report is included in full. Table and chart numbers and references have been updated for this document. Readers can find the original report here.

The NYS Public Service Commission is increasingly focused on investing in strengthening the resiliency of the electricity grid. At the local level, understanding the frequency, duration, and causes of electrical outages is crucial for anticipating potential future outages. To this end, outage data provided by the electric utility, New York State Electric and Gas (NYSEG) for the last three years in Tompkins County is presented above.

Table 1 shows the number of electrical outages in Tompkins County from 2020 to 2023. NYSEG tracks the causes for each outage. Those detailed descriptions were grouped into six categories: vegetation interference, equipment failure, animal interference, maintenance, weather, and miscellaneous. It is clear the largest number of outages arise from vegetative interference, such as a tree or branch falling on a transmission line. Equipment failures account for 25.5% of all outages and could include anything from defective switch to transformer damage. The Miscellaneous category includes outages

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https://www.jec.senate.gov/public/index.cfm/democrats/2024/1/how-renewable-energy-can-make-the-power-grid-more-reliable-and-address-risks-to-electricity-infrastructure

[162] Martin, L. and Brehm, K. *Clean Energy 101: Virtual Power Plants* (2023). Rocky Mountain Institute. https://rmi.org/clean-energy-101-virtual-power-plants/

[163] Siegner, K., et al. *Cheaper, Cleaner, Faster* (2023). Rocky Mountain Institute. https://rmi.org/cheaper-cleaner-faster/

caused by installation of cable lines, adding new customer loads, or loss of supply[164] . The outages totaled 2,236 over the 1,095 days (three years) in the data set reviewed. This averages at two outages per day in Tompkins County over the last three years.

Table 1: Number of Electrical Outages in Tompkins County by Cause from 2020-2023		
Cause of Outage	**Number of Outages**	**Percent of Outages by Cause**
Vegetation Interference	1,210	54.11%
Equipment Failure	571	25.54%
Animal Interference	156	6.98%
Maintenance	129	5.77%
Weather	104	4.65%
Miscellaneous	66	2.95%
Total Number of Outages	**2,236**	

Beyond the number of outages, another important consideration is the duration of those outages. Table 2 shows the Outage Duration Statistics for Tompkins County from 2020 to 2023. The minimum outage in Tompkins County was ten minutes, with the longest outage lasting fourteen hours and forty-three minutes. The average outage duration was 3 hours and 20 minutes. Understanding the duration of outages will help tailor short-term or long-term solutions for grid outages.

Table 2: Outage Duration Statistics in Tompkins County	
	Hours
Minimum	0.17
Average	3.33
Maximum	14.71

Grid Vulnerabilities & Strengths

As electrification is increasingly adopted in New York State, the State and utility partners are implementing steps to address electric grid vulnerabilities and strengths to reduce the likelihood of outages. This section outlines some of the major issues being studied and addressed by those partners.

Tompkins County and other Upstate New York communities may be somewhat better insulated from future brownouts due to their relatively lower electric load compared to Downstate areas. Additionally, the Upstate grid's large amount of hydroelectric power lends some stability to the electric supply in Upstate New York[165]. However, the

[164] NYSEG provided all the outage data, which is publicly available. For each of the following combination: incident number + date + circuit, the entry is unique with only the longest outage being counted.
[165] New York Independent Systems Operator (NYISO). *Real Time Data Dashboard: Real-Time Fuel Mix and Load with Losses*. https://www.nyiso.com/real-time-dashboard

intermittent nature of wind and solar energy, which supplement hydroelectric power as the grid moves towards more renewable energy sources, presents challenges. Solar generation peaks during the day and during the summer, and declines before evening loads rise, while wind generation is greater at night when demand is typically lower. In the winter, there is generally lower solar generation from limited daylight, which must be considered as cold weather impacts EV battery consumption and capacity and demands for electrified building heating systems increase. Understanding these nuances and how to manage generation versus demand will be key to reducing stress on the grid as electrification grows.

Climate change and the strengthening of storms and extreme temperatures can also further exacerbate grid vulnerabilities. Renewable energy power shortages are likely to arise if there are co-occurring high temperatures with wind or water droughts in the summer or lower temperatures with wind droughts in the winter[166] . Similarly, higher temperatures lead to higher energy demands for cooling, which strains the grid. This is more likely as the planet's temperature is expected to rise. Similarly, severe storms or increased instances of drought can impact infrastructure that's already running at capacity. In an electrified future, the amount of electric energy needed will increase because the transportation and building sectors will increasingly be requiring electricity to power vehicles and heating loads. The change in electric energy demand will result in New York State transitioning from a summer-peaking to a winter-peaking region. This introduces additional challenges for a future where electrification is standard. Peaking is the time when the electric grid has the highest electricity demand systemwide, and it introduces vulnerabilities to outages if supply cannot keep pace with demand. Current summer-peaking typically occurs on high-temperature days when demand for cooling is highest. As building HVAC systems move from fossil fuel to heat pumps, the resulting increase in electricity usage for heating is expected to reconfigure the system so that the highest electricity demand occurs in winter months during the coldest hours of the year. This is a concern as the risks for loss of heat in the winter are significant.

As the grid becomes increasingly "smart," it also provides more openings for bad actors to access the system and thus becomes more susceptible to cyber attacks. Cyber attacks to the grid could disrupt service and cause outages. The Department of Energy's Office of Cybersecurity, Energy Security, and Emergency Response (CESER) has seen significant

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[166] Liu, M. V., Cornell University Systems Engineering Department. "Heterogeneous Vulnerability of Zero-Carbon Power Grids Under Climate-Technological Changes." Cornell University, Ithaca, NY, December 16, 2024. https://arxiv.org/pdf/2307.15079.

investments from the federal government to help reduce cyber risks and strengthen the resilience of the grid[167] .

[167] Office of Cybersecurity, Energy Security, and Emergency Response (CESER). "DOE Announces $45 Million to Protect Americans From Cyber Threats and Improve Cybersecurity in America's Energy Sector". U.S. Department of Energy. February 26, 2024. https://www.energy.gov/articles/doe-announces-45-million-protect-americans-cyber-threats-and-improve-cybersecurity

RECOMMENDATIONS

CAP RECOMMENDATIONS

Icon Key

ICON IS FULL COLOR

Recommended action benefits the sector the icon represents. E.g. this action benefits labor

ICON IS GREY

Recommended action does not directly benefit the sector the icon represents. E.g. this action does not benefit housing

EMISSIONS IMPACT

None

Low

Medium

High

RESILIENCE IMPACT

None

Low

Medium

High

TIMELINE

Short-term

<1 year

Medium-term

1-3 years

Long-term

4-5 years

2030 CLIMATE ACTION PLAN

HOUSING

safety • efficiency • affordability • accessibility

HOUSING

safety • efficiency • affordability • accessibility

DRAFT

	Action	Impact	Timeline
1	***Promote Density & Infill Housing*** *Identify ways to promote and incentivize density and infill housing to increase supply and decrease greenhouse gas emissions per housing unit.*		
2	**Participatory Housing Process** *Utilize a participatory process to identify solutions to homelessness by including those with lived experience.*		
3	**Implement Home Energy Score Ordinance** *Implement an ordinance requiring residential properties receive a Residential Energy Score and require score disclosure at lease and sale of property.*		
4	**Transportation Access Minimums** *Identify municipal codes that could be locally adopted requiring a minimum distance between new residential construction and public transportation (bikeshare hubs, TCAT, Carshare).*		

HOUSING

safety • efficiency • affordability • accessibility

DRAFT

	Action	Impact	Timing
5	**Portable Heat Pump Pilot** *Collaborate with manufacturers and Tompkins Whole Health to purchase and distribute portable heat pumps to residents of rental properties.*		
6	**Small-Scale Solar & Batteries** *Collaborate with manufacturers and Tompkins Whole health to distribute small-scale solar panels and batteries to residents with electricity-dependent medical equipment.*		
7	**Flood Resilient Landscaping & Surfaces** *Incentivize flood resilient landscaping and permeable surfaces in all new commercial and multifamily residential development*		
8	**Rent Stabilization** Identify and implement opportunities for rent stabilization.		

HOUSING

safety • efficiency • affordability • accessibility

DRAFT

	Action	Impact	Timing
9	**Commercial Building Performance Standards** *Implement a phase program that require benchmarking, reporting, and achievement of minimum energy efficiency standards for all commercial buildings.*		
10	**Explore Local NFIP Subsidies** *Explore opportunities and funding sources to locally subsidize the cost of the National Flood Insurance Program for property owners.*		

2030 CLIMATE ACTION PLAN

LABOR

accessibility • stability • growth • opportunity

LABOR

accessibility • stability • growth • opportunity

DRAFT

	Action	Impact	Timing
1	**Ban the Box** *Prohibit employers within the city, where legal, from conducting criminal background checks or asking applicants if they have a criminal history prior to offering an applicant conditional employment.*	CO₂	
2	**Acclimation & Rest Requirements** *Require acclimation of new and returning workers to hot conditions over 7-14 days to prevent heat-related illness. Require employers to provide access to cool, potable water at all times and 15-min rest breaks every two hours for employees working in temperatures at or above 90ºF.*	CO₂	
3	**Just Cause Employment** *Collaborate with Tompkins County and external partners to identify and adopt legislation that would protect workers against unjustified firings and dismissals.*	CO₂	
4	**Mobile Rehabilitation & Command Vehicle(s)** *Utilize state and federal grants to subsidize the investment in a mobile rehabilitation vehicle, or units that provide critical support (HVAC, restrooms, medical equipment, etc.) to first responders during extended duration emergency events.*	CO₂	

LABOR

accessibility • stability • growth • opportunity

DRAFT

	Action	Impact	Timing
5	**Reskilling & Upskilling Training** *Invest in reskilling and upskilling training programs for City employees whose jobs may be impacted by the energy transition. Provide opportunities for employees to learn transferable or new, marketable skills. E.g. Electric vehicle maintenance.*		
6	**Invest in Bus Networks** *Invest in networks supporting public transit, like dedicated lanes, busways, and signal priority, to ensure reliable, timely access to destinations, including employment, training, and interviews.*		
7	**Support Workforce Development Programs** *Provide financial and/or administrative support to workforce development programs that primarily serve City residents.*		
8	**Assist with Securing a Downtown Training Center** *Facilitate the identification and lease/purchase of a downtown workforce training center to be used by local training partners.*		

LABOR

accessibility • stability • growth • opportunity

	Action	Impact	Timing
9	**Living Wage** *Collaborate with Tompkins County and external partners to identify pathways toward a living wage ordinance.*		

2030 CLIMATE ACTION PLAN

PUBLIC HEALTH

prevention • respect • compassion • accessibility

PUBLIC HEALTH

prevention • respect • compassion • accessibility

DRAFT

	Action	Impact	Timing

1

Air Quality Monitoring & Dashboard

Utilize state an federal grants to deploy air quality monitoring devices and a public dashboard to track common pollutants. E.g. PM2.5, SOx, NOx,.

2

Identify Active Transportation Corridors

Identify locations for active transportation corridors that prohibit the use of internal combustion engines and prioritize micromobility. Once identified, launch a pilot season and report impacts.

3

Deploy Additional Cooling/Warming Centers

Identify and deploy additional low and no-barrier cooling and warming centers for Ithaca's most challenging outdoor seasons.

4

Maximum Temperature Ordinance

Work with public health and building energy experts to determine a maximum safe indoor temperature threshold and propose new maximum temperature ordinance for City rental housing.

PUBLIC HEALTH

prevention • respect • compassion • accessibility

DRAFT

	Action	Impact	Timing
5	**CDC BRACE Partnership** *Convene appropriate partners to participate in the CDC Building Resilience Against Climate Effects (BRACE) framework process to identify public health risks and solutions.*		
6	**Support & Expand Induction Cooktop Program** *Support City resident enrollment and expand Tompkins County's induction cooktop pilot program, which provides free, single-burner cooktops to County residents.*		
7	**Support & Expand City Forestry Program** *Support the expansion of the City's robust street tree and forest management program to increase the number of new, and protection of existing, shade trees.*		
8	**Age-Friendly Community** In collaboration with appropriate partner agencies, utilize the World Health Organization Age-Friendly Cities framework to identify service and resource gaps in the community.		

PUBLIC HEALTH

prevention • respect • compassion • accessibility

DRAFT

	Action	Impact	Timing
9	**Mobile Rehabilitation & Charging Vehicles** *In partner coalitions, invest in mobile rehabilitation and charging vehicles to provide onsite health and health equipment support to people experiencing homelessness or in the event of a sustained power outage.*		
10	**Employer Incentivized Micromobility** *Identify ways to incentivize local employees or employers to utilize micromobility or public transit in their commute.*		
11	**Vector Disease Campaigns** *Partner with Tompkins Whole Health to launch a climate and public health-focused vector borne disease education campaign.*		
12	**Waste Reduction** *Explore and identify opportunities to collaborate with necessary stakeholders to reduce waste and packaging.*		

2030 CLIMATE ACTION PLAN

RACIAL EQUITY

leadership • accountability • transparency • intention

RACIAL EQUITY

leadership • accountability • transparency • intention

DRAFT

Action | Impact | Timing

1

CJC Survey & Mapping

Complete robust climate justice community survey followed by data mapping to ensure Justice50 budgeting goals are directed to appropriate neighborhoods.

2

Position BIPOC Communities as Leaders

Design new resilience and adaptation programs so that BIPOC communities and community centers lead educational projects.

3

Balance Representation

Ensure that committees, working groups, and commissions are racially and economically representative of the communities they serve.

4

Robust Community Engagement

Develop a robust community engagement strategy that utilizes a variety of mediums, including art, music, and other non-traditional communications.

RACIAL EQUITY

leadership • accountability • transparency • intention

DRAFT

	Action	Impact	Timing

5 **Prioritize CJC for Investment**

Using Justice50 principles, prioritize climate justice communities for infrastructure and other sustainability investments.

6 **Support Transitional Employment**

Identify and deploy strategies to support transitional employment between job training programs and permanent career placement.

7 **Identify Entrepreneurial Training Partner**

Identify skilled entrepreneurial training partner(s) to assist with business-ownership training for individuals in workforce development programs.

8 **Community-Wide Priority Survey**

Deploy a community-wide priority survey and respect and support the results, prioritizing activities mentioned by low-income and communities of color.

RACIAL EQUITY

leadership • accountability • transparency • intention

DRAFT

	Action	Impact	Timing
9	**Develop an Equity Screening Tool** *Develop and deploy an equity screening tool in partnership with community leaders of color to help gauge impacts of policy decisions on vulnerable populations.*	CO_2	
10	**Join Resilient Cities Network** *Assess the value of joining the Resilient Cities Network, a coalition of municipalities that work together to identify and introduce new ways of building cities from an equity perspective.*	CO_2	
11	**Develop a Racial Equity Action Plan** *Under the leadership of, and in partnership with, organizations of color, develop a Racial Equity Action Plan for the City of Ithaca.*	CO_2	
12	**Routinely Reassess Policies & Procedures** *Identify a routine cycle for policy and procedure reassessment to ensure the burden of impact is not felt most harshly on low-income and communities of color.*	CO_2	

2030 CLIMATE ACTION PLAN

EMERGENCY RESPONSE

resilient • proactive • collaborative • comprehensive

EMERGENCY

resilient • proactive • collaborative • comprehensive

	Action	Impact	Timing
1	**Hire an Emergency Manager** *Secure funding for and hire an Emergency Manager to produce emergency plans, facilitate emergency communications, and build critical partnerships for emergency resource planning.*		
2	**Develop General Evacuation Plans** *Develop robust and comprehensive evacuation plans that are accessible in multiple languages and by people with varying abilities.*		
3	**Identify Responder Equipment Needs** *Identify needs and gaps in necessary equipment and response vehicles within IPD and IFD, including high-water rescue vehicles, command vehicles, and PPE.*		
4	**Develop Vulnerable Population Evacuation & Response Plans** *In collaboration with vulnerable communities, produce emergency evacuation and response plans that consider the specific needs of particularly vulnerable populations, including individuals with disabilities, the homeless, and very low-income families.*		

EMERGENCY resilient • proactive • collaborative • comprehensive

	Action	Impact / Timing
5	**Invest in Senior Leadership Training** *Invest in and deploy emergency training for all senior City leadership, including communication strategies, roles, responsibilities, and availability of resources.*	
6	**Deploy Community Resiliency Training** *Invest in and deploy community-led resiliency training workshops including how to protect private property, emergency preparedness, and building social networks and capacity.*	
7	**Develop Extreme Heat & AQ Plan** *Develop and deploy a plan for extreme heat or poor air quality for individuals experiencing homelessness. This may be a plan similar to Code Blue.*	
8	**Create a Resiliency Committee** *Identify necessary stakeholders and create a resiliency committee consisting of City staff, external partners, and service providers to begin creating a robust and connected network of resources.*	

EMERGENCY

resilient • proactive • collaborative • comprehensive

	Action	Impact	Timing
9	**Pursue Emergency Management Accreditation** *Pursue New York State Local Emergency Management Accreditation through the Department of Homeland Security and Emergency Services.*		
10	**Invest in Green Infrastructure** Robustly invest in green infrastructure and bio-solutions to mitigate flood hazards.		
11	**Identify Resilience Hubs** *Identify and invest in resilience hubs throughout the City that can serve as an emergency center, heating and cooling center, and planning resource for residents.*		
12	**Deploy EV Charging Along Evacuation Routes** *Ensure adequate electric vehicle charging, including DC fast charging, is available along evacuation routes and is accessible to emergency vehicles.*		

EMERGENCY

resilient • proactive • collaborative • comprehensive

	Action	Impact	Timing
13	**Identify Emergency Medical Response Plan** *In partnership with local first responders, identify an emergency medical response plan in the event that roads are impassable due to a flood.*		
14	**Identify Telecommunications Gaps** Perform an essential services telecommunications gap analysis.		

2030 CLIMATE ACTION PLAN

POWER RELIABILITY

affordable • renewable • local • resilient

POWER

affordable • renewable • local • resilient

	Action	Impact / Timing
1	**Decentralize Power Generation** *Secure funding for and deploy decentralized power generation, including utility-scale and small-scale renewables.*	
2	**Rate Case Participation** *Participate in the tri-annual utility rate case and advocate for distribution infrastructure resilience, affordability, and decarbonization consistent with CLCPA goals.*	
3	**Heat Pump Rate Advocacy** *Advocate to the NYS PSC to pilot a Tompkins County heat pump winter electric rate for the NYSEG service territory.*	
4	**Aggressive Deployment of DERs** *Rapid and aggressive deployment of publicly- and City-owned renewable energy systems, energy storage devices, and other grid-interactive, distributed energy resources (DERs).*	

POWER

affordable • renewable • local • resilient

	Action	Impact	Timing
5	**Parking Lot Solar Ordinance** *Maximize low-value land value by requiring canopy solar project development in under-utilized and oversized parking lots.*		
6	**Load Control through VPPs** *Increase electric reliability, resilience, and stability by aggregating and coordinating distributed energy resources through virtual power plants.*		
7	**Pursue Autonomous Microgrid(s)** *Pursue long term local energy independence, decentralization, and resilience through the development of autonomous microgrids, or smaller scale, localized electrical grids capable of generating, storing, and distributing power through AI.*		
8	**End-of-Life Equipment Ordinance** *Mandated replacement of fossil fuel equipment with electric alternatives at end of useful life to decrease necessary utility investment in natural gas infrastructure.*		

POWER

affordable • renewable • local • resilient

Action | Impact | Timing

9

Deployment of Weatherization Teams

Invest in the wide-scale deployment of low and no-cost residential weatherization teams across the City.

10

Benchmarking & Performance Standards

Decrease total power load by requiring commercial energy benchmarking and minimum energy performance standards for all existing commercial buildings, including City facilities.

11

Aggregate Rooftop Solar Contracts

Using digital twin technology, aggregate potential building portfolios ideal for rooftop solar installation and aggregate into solar contract portfolios to maximize cost savings and energy yield.

12

Strategic Charging & Generation Hubs

Strategically deploy shared-service EV charging hubs that utilize clean electricity generation, storage, and bidirectional, grid-interactive charging.

Climate Action Plan Recommendations Matrix

NO.	Recommendation	Page #	Housing	Labor	Public Health	Equity	Emergency Response	Power Reliability	GHG Reduction	Resilience	Project Start Timeline	Projected Cost to City
	Housing											
1	Promote Density & Infill Housing		☑	☑	☑	☐	☐	☑	███	███	Short	$
2	Participatory Housing Process		☑	☐	☑	☑	☐	☐		██	Short	$
3	Home Energy Score Ordinance		☑	☑	☑	☐	☐	☑	███	█	Medium	$
4	Transportation Access Minimums		☑	☑	☑	☑	☐	☐	███	█	Medium	$
5	Portable Heat Pump Pilot		☑	☐	☑	☑	☐	☑	█	█	Medium	$$
6	Small-Scale Solar & Batteries		☑	☐	☑	☑	☑	☑	█	███	Long	$$
7	Flood Resilient Landscaping & Surfaces		☑	☑	☑	☐	☑	☐	█	███	Short	$
8	Rent Stabilization		☑	☑	☑	☑	☐	☐		█	Short	$
9	Commercial Building Performance Standards		☑	☑	☑	☐	☐	☑	████	█	Short	$$$
10	Local NFIP Subsidies		☑	☐	☐	☑	☐	☐		███	Long	$$$
	Labor											
1	Ban the Box		☑	☑	☑	☑	☐	☐		█	Short	$
2	Acclimatization & Rest Requirements		☐	☑	☑	☑	☑	☐		███	Medium	$
3	Adopt Just Cause Employment		☑	☑	☑	☑	☐	☐		█	Short	$
4	Mobile Rehabilitation & Command Vehicles		☐	☑	☑	☐	☑	☑	███	███	Long	$$$
5	Reskilling & Upskilling Training		☑	☑	☑	☑	☐	☐	█	█	Medium	$$
6	Invest in Bus Networks		☑	☑	☑	☑	☑	☐	████	█	Long	$$
7	Support Workforce Development Programs		☑	☑	☑	☑	☐	☐	█	███	Short	$$
8	Assist with Securing a Downtown Training Center		☑	☑	☑	☑	☐	☐	███	███	Short	$
9	Adopt a Local Living Wage		☑	☑	☑	☑	☐	☐		█	Short	$
	Public Health											
1	Air Quality Monitoring Dashboard		☑	☐	☑	☑	☐	☐	█	███	Medium	$$
2	Identify Active Transportation Corridors		☑	☑	☑	☑	☑	☐	████	█	Long	$$
3	Deploy Additional Cooling/Warming Centers		☑	☑	☑	☑	☑	☐	█	███	Short	$$
4	Maximum Temperature Ordinance		☑	☑	☑	☑	☐	☐	███	███	Medium	$
5	CDC BRACE Partnership		☑	☑	☑	☑	☑	☐	██	█	Short	$
6	Support & Expand Induction Cooktop Program		☑	☐	☑	☑	☐	☐	█	█	Medium	$
7	Support & Expand City Forestry Program		☐	☐	☑	☑	☐	☑	██	███	Short	$$
8	Age-Friendly Community		☑	☑	☑	☑	☐	☐	█	███	Short	$
9	Mobile Rehabilitation & Charging Vehicles		☑	☑	☑	☑	☑	☑	██	██	Long	$$$
10	Employer Incentivized Micromobility		☑	☑	☑	☑	☑	☐	████	█	Medium	$
11	Vector Disease Campaigns		☑	☑	☑	☑	☐	☐	█	███	Short	$
12	Waste Reduction		☐	☐	☑	☑	☐	☐	▌	▌	Short	$
	Racial Equity											
1	CJC Survey & Mapping		☑	☑	☑	☑	☐	☐	█	█	Short	$
2	Position BIPOC Communities as Leaders		☑	☑	☑	☑	☑	☐	██	██	Short	$
3	Balance Representation		☑	☑	☑	☑	☐	☐	█	███	Short	$
4	Robust Community Engagement		☑	☑	☑	☑	☑	☑	███	███	Short	$
5	Prioritize CJC for Investment		☑	☑	☑	☑	☐	☐	█	█	Short	$
6	Support Transitional Employment		☑	☑	☑	☑	☐	☐	████	██	Medium	$
7	Identify Entrepreneurial Training Partner		☑	☑	☑	☑	☐	☐	█	███	Medium	$

8	Community-Wide Priority Survey		☑	☑	☑	☑	☑	☑			Medium	$
9	Develop an Equity Screening Tool		☑	☑	☑	☑	☐	☐			Short	$
10	Join Resilient Cities Network		☑	☑	☑	☑	☐	☐			Short	$
11	Develop a Racial Equity Action Plan		☑	☑	☑	☑	☐	☐			Medium	$
12	Routinely Reassess Policies & Procedures		☑	☑	☑	☑	☐	☐			Short	$
Emergency Response												
1	Hire an Emergency Manager		☑	☐	☑	☑	☑	☐			Short	$$
2	Develop General Evacuation Plans		☑	☑	☑	☑	☑	☐			Short	$
3	Identify & Invest In Responder Equipment Needs		☐	☑	☑	☐	☑	☐			Medium	$$
4	Develop Vulnerable Population Evacuation & Response Plan		☐	☐	☑	☑	☑	☐			Short	$
5	Invest in Senior Leadership Training		☐	☑	☑	☑	☑	☐			Medium	$$
6	Deploy Community Resilience Training		☑	☑	☑	☑	☑	☑			Long	$
7	Develop Extreme Heat & AQ Plan		☑	☑	☑	☑	☐	☐			Medium	$
8	Create a Resilience Committee		☑	☑	☑	☑	☐	☐			Short	$
9	Pursue Emergency Management Accreditation		☑	☑	☑	☑	☐	☐			Long	$
10	Invest in Green Infrastructure		☑	☑	☑	☑	☑	☐			Long	$$$
11	Identify Resilience Hubs		☑	☑	☑	☑	☐	☐			Medium	$
12	Deploy EV Charging Along Evacuation Routes		☑	☑	☑	☑	☑	☐			Long	$
13	Identify Emergency Medical Response Plan		☑	☑	☑	☑	☐	☐			Short	$
14	Identify Telecommunications Gaps		☑	☑	☑	☑	☑	☐			Short	$
Power Reliability												
1	Decentralize Power Generation		☑	☑	☑	☑	☐	☑			Long	$
2	Rate Case Participation		☑	☐	☑	☑	☑	☑			Short	$$
3	Heat Pump Rate Advocacy		☑	☑	☑	☑	☐	☑			Short	$
4	Aggressive Deployment of DERs		☑	☑	☑	☑	☐	☑			Long	$$$
5	Parking Lot Solar Ordinance		☑	☑	☑	☐	☐	☑			Short	$
6	Load Control Through VPPs		☑	☐	☑	☐	☑	☑			Long	$$$
7	Pursue Autonomous Microgrid(s)		☑	☐	☑	☑	☑	☑			Long	$$$
8	End-of-Life Equipment Ordinance		☑	☑	☑	☑	☐	☑			Medium	$
9	Deployment of Weatherization Teams		☑	☑	☑	☑	☐	☑			Medium	$
10	Benchmarking & Performance Standards		☑	☑	☑	☑	☑	☑			Medium	$
11	Aggregate Rooftop Solar Contracts		☑	☑	☑	☑	☐	☑			Long	$
12	Strategic Charging & Generation Hubs		☑	☑	☑	☑	☐	☑			Long	$$$